USER'S MANUAL

S-100 VIEWER

Prepared for: International Hydrographic Organization (IHO)

Developed by: United States Navy (Naval Information Warfare Center Atlantic) in support of the S-100 Working Group

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2	09/01/2016	Revised for S-100 Viewer - Version 1.2.0.0
3	03/06/2017	Revised for S-100 Viewer - Version 1.3.0.0
4	08/24/2020	Revised for S-100 Viewer - Version 1.9.0.0
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USER'S MANUAL

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1.0 GENERAL INFORMATION

1.0 GENERAL INFORMATION

The S-100 Viewer provides a mechanism to better understand the portrayal requirements of S-100 based products and verifies that those requirements are properly partitioned and implemented. The portrayal displayed in the viewer provides verification that the specifications define a clear, unambiguous process for translating from a products dataset encoding to a portrayal via automated generation of portrayal instructions.

1.1 System Overview

The S-100 Viewer is a tool for evaluating S-100 data sets, feature catalogs, and portrayal catalogs. The current version is primarily focused on loading S-101 Electronic Navigational Charts (ENC) datasets, with best-effort support for GML and HDF-5 product encodings. The viewer is a realization of Test Bed Phase 3 as shown in the S-100 System Overview.

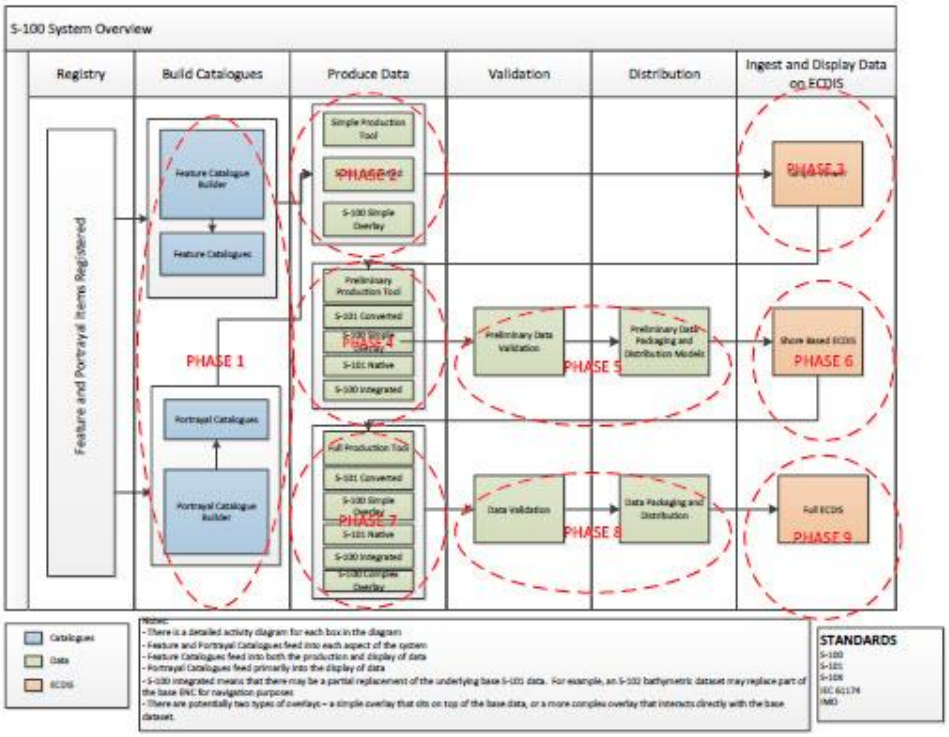


Figure 1. S-100 System Overview

The software consists of a stand-alone Windows application that does not require network connectivity.

Current Support Limitations of version 2.0.0:

Product	FC Versions	PC Versions
S-101	v1.0.2	v1.0.2
	v.1.1.0	v1.1.0
	v.1.2.3	v1.2.3
	v.1.5.0	v1.5.0
	v.2.0.0	v.2.0.0
S-102	v.2.0.0	v.1.0.0
	v.2.1.2	v.1.0.0
	v.2.2.0	v.1.0.0
	v.3.0.0	v.3.0.0
S-104	v.2.0.0	n/a
S-111	v.1.0.0	v.1.0.0
	v.1.2.0	v.1.0.0
	v.2.0.0	v.2.0.0
S-124	v1.0.0	v1.0.0
	v2.0.0	v2.0.0
S-129	v1.0.0	v1.0.0
	v1.1.0	v1.1.0
	v2.0.0	v2.0.0

1.2 Project References

S-100 Universal Hydrographic Data Model, Edition 5.2.0
S-101 ENC Product Specification, Edition 2.0.0

1.3 Points of Contact

For assistance, please contact our development team: NIWCTestbed@us.navy.mil

1.4 Organization of the Manual

General Information (Section 1)
A brief overview of the S-100 Viewer along with primary points of contact.

System Summary (Section 2)
Basic system configuration and summary of major data flow.

Getting Started (Section 3)
Quick start guide.

Using S-100 Viewer (Section 4)
More detailed guide.

Presentation Browsers (Section 5)
Data presentation browsers and their functionality.

Chart Area (Section 6)
The S-100 Viewer Chart Area.

Log Panel (Section 7)
The S-100 Viewer diagnostic information for monitoring errors, warnings and performance information.

2.0 SYSTEM SUMMARY

2.0 SYSTEM SUMMARY

S-100 Viewer has been developed to run on Windows 10 or 11. System configurations from laptops to high-performance desktop computing environments are supported.

2.1 System Configuration

The S-100 Viewer application is a stand-alone application and does not require a network connection as long as access to S-100 data is available on local storage. User input consists of keyboard and mouse and output is provided on a single monitor.

2.2 Data Flows

The current version of the S-100 Viewer allows for S-100 data sets and S-100 feature catalogues to be loaded into system memory for review and analysis.

3.0 GETTING STARTED

3.0 GETTING STARTED

The S-100 Viewer allows for review of S-100 Feature Catalogues and data set files. It provides both tabular and graphical displays. Figure 2 shows the S-100 Viewer after an S-101 data file has been loaded, and "Portrayal Tab" has been modified to turn off some of the Viewing Group Layers.

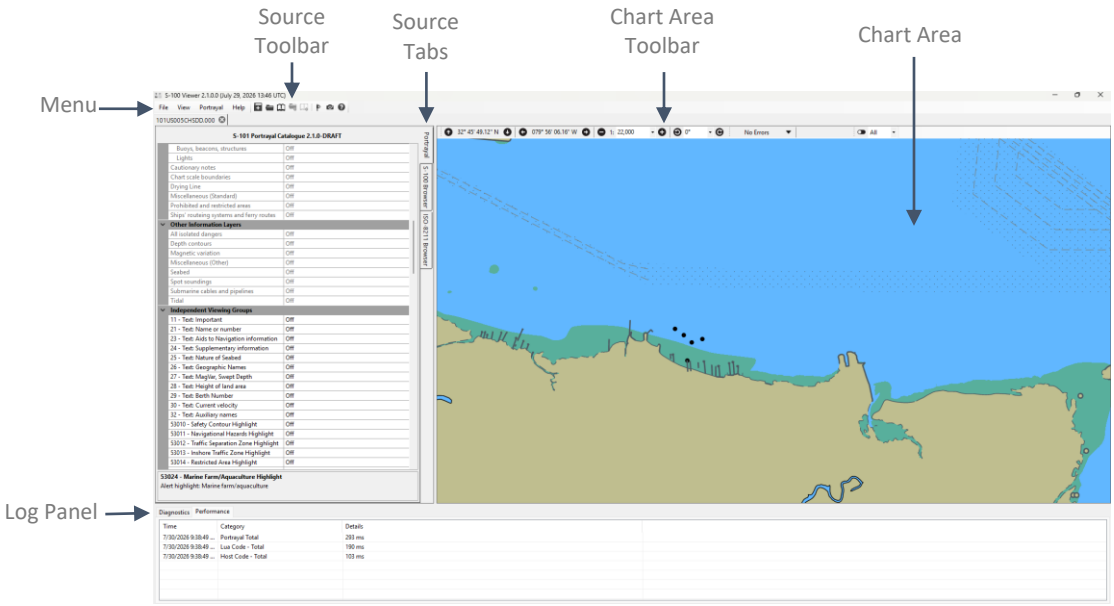


Figure 2. S-100 Viewer User Interface

3.1 S-100 Viewer Installation

The S-100 Viewer installation process does not require special configuration. Once the product is installed a shortcut is available via the "Start" menu. Figure 3 depicts the initial screen of the S-100 Viewer installation program. The installation program is self-guided; it is recommended that the defaults are accepted on each screen.

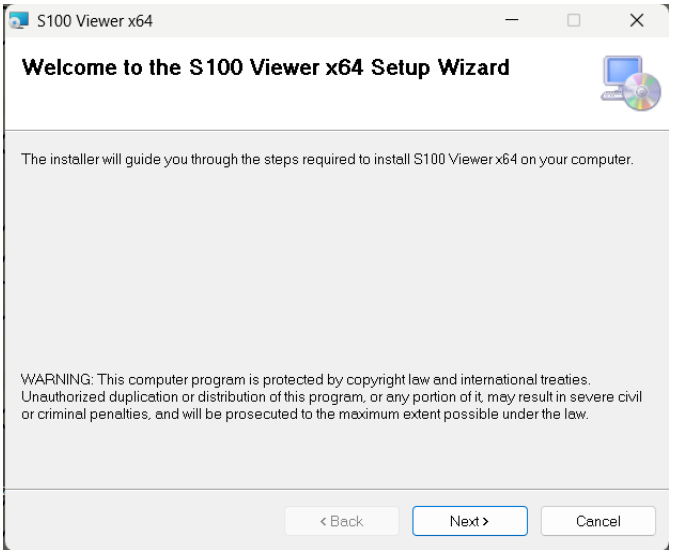


Figure 3. S-100 Viewer installation program.

3.1.1 Installation Location

During installation, a dialog will prompt for the installation location for the S-100 Viewer application. A default file path will be provided. If an alternate installation path is required, then select the appropriate path using the “Browse” button. Choose Install S100 Viewer x64 for Everyone who uses this computer. Select “Next >” to continue. Then select “Next >” on the “Confirm Installation” step.

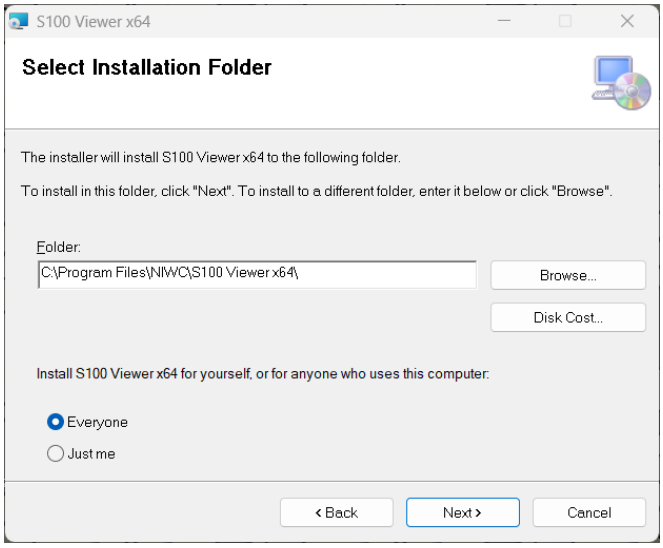


Figure 4. Installation Location

3.1.2 Installer Request Permission to modify Computer

Allow the installer to modify machine. Windows User Account Control will display a request for admin privileges asking “Do you want to allow this app from an unknown publisher to modify your device?”. Accept by clicking “Yes”.

3.1.3 Installation Complete

When the installation process is complete, a dialog window will be presented After installation is complete you can find the “S100 Viewer x64” program in your applications in the windows start menu, or at the install location from “Select Installation Folder” step.

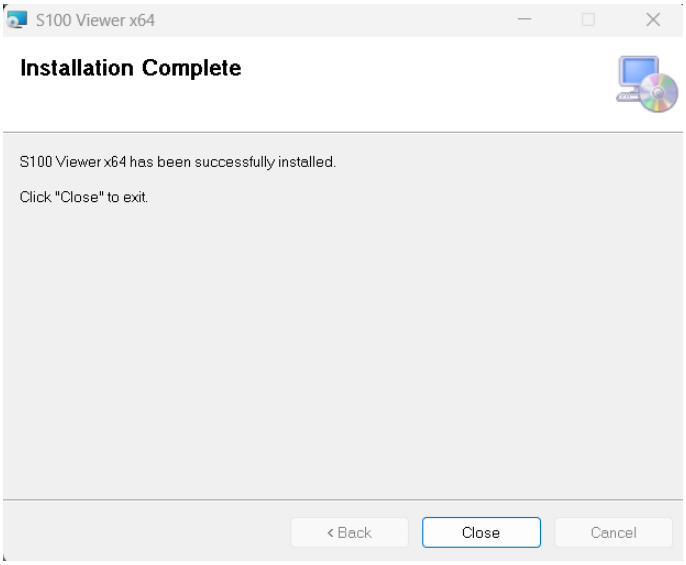


Figure 5. Installation Complete

3.1.4 Registration

The first time the program runs, or any time it is run without a valid product key registered for the install, you will see the “NIWC S100Viewer 2.1.0.0 Registration Form”. In order to register the application fill out the form fields, and click “Generate Email”. If this button is able to find the default email client for your system, it should prepare an email to NIWCTestbed@us.navy.mil with the required details to register the application. If it does not find the proper default email client for your system, you may need to use “Copy Registration Info” button, and copy the required information to send an email to register the application.

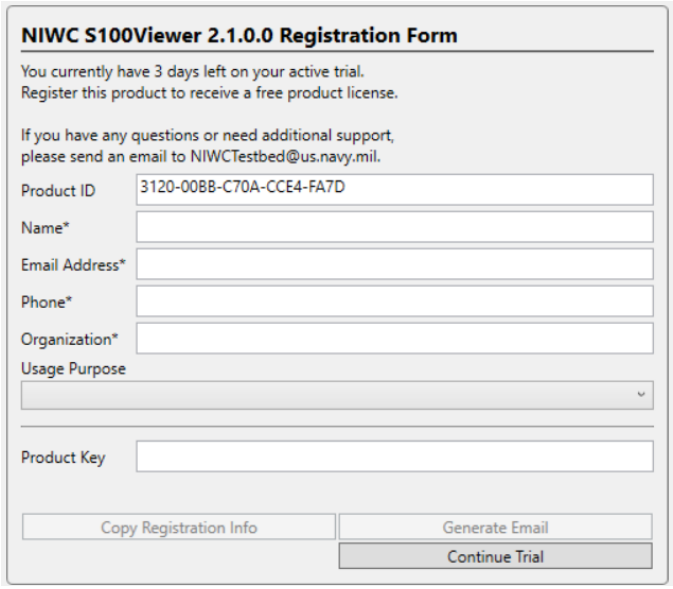


Figure 6. Registration Form

4.0 USING S-100 VIEWER

4.0 USING S-100 VIEWER

This section provides a detailed description of S-100 Viewer menu entries and functions. Figure 7 identifies the necessary user control items.

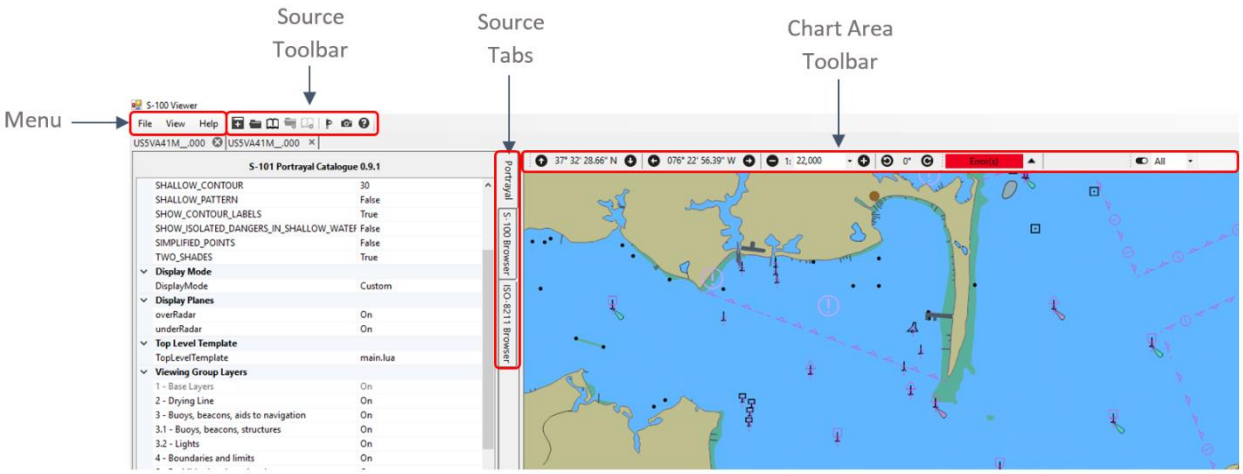


Figure 7. S-100 Viewer User Controls

4.1 Menu Bar

The Menu Bar is shown below in Figure 8. This section details what is available in each menu item.

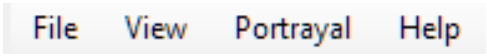


Figure 8. Menu Bar Layout

4.1.1 File Menu

The "File" Menu is used to open S-100 data set files and feature catalogues, take screen shots, and exit the S-100 Viewer. The File menu is shown in Figure 9.

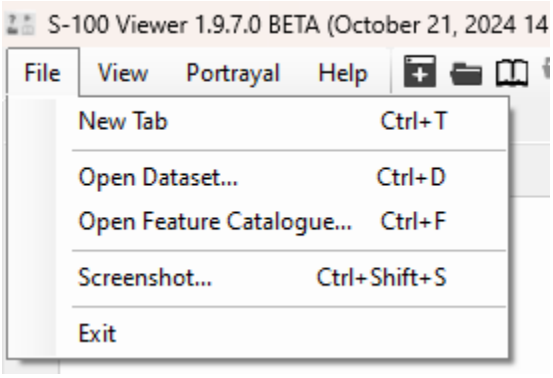


Figure 9. File Menu Options

4.1.1.1 New Tab

The “File → New Tab” option creates a new tab as seen in Figure 10. This duplicates the tab shown on startup.

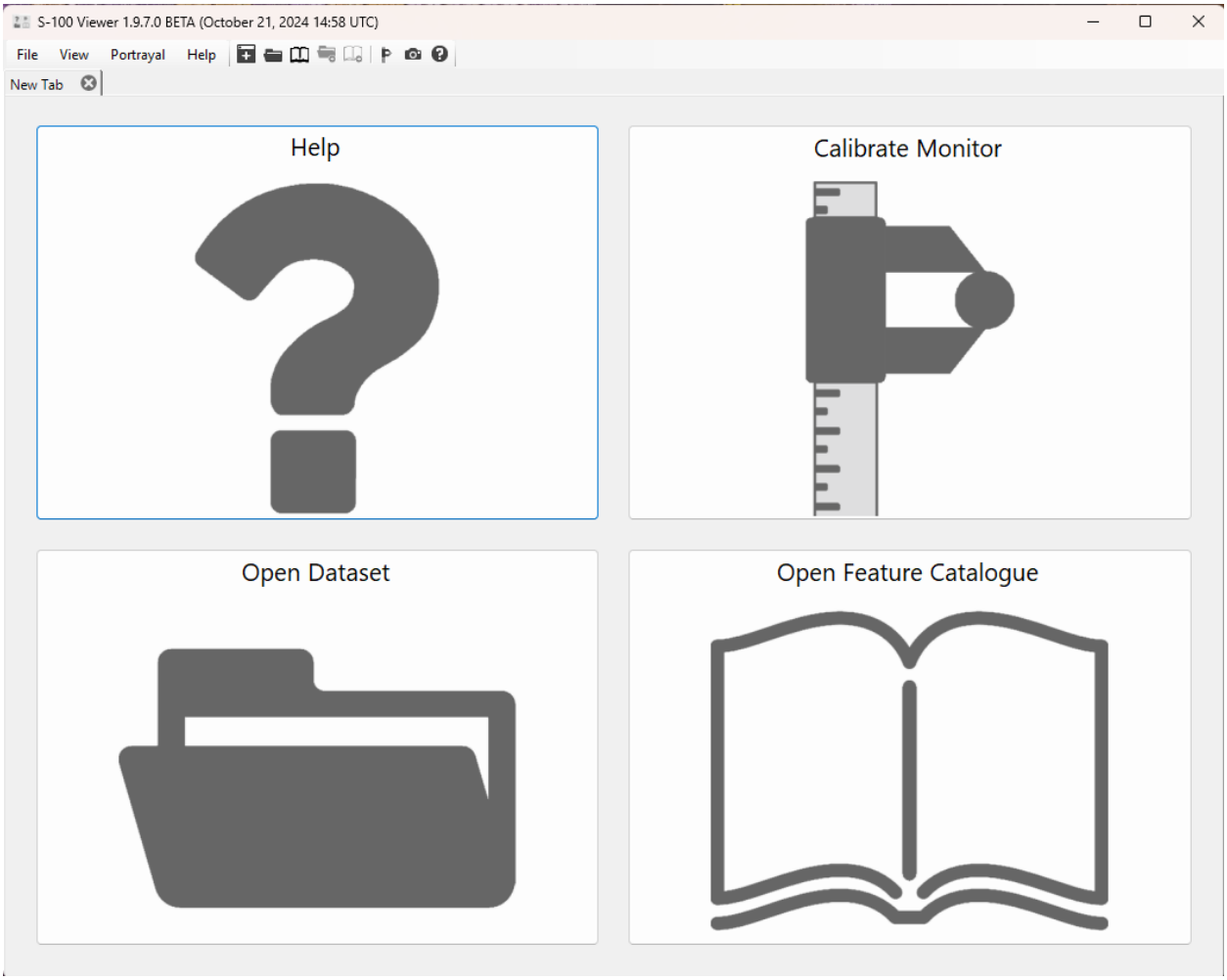


Figure 10. New Tab

4.1.1.2 Open Dataset

The S-100 Viewer supports multiple browsing modes to inspect an S-100 data set file. The browsing mode is determined by whether a feature and/or portrayal catalog are selected during the file opening procedure.

The S-100 Viewer also supports multiple methods of opening datasets. By clicking “Open Dataset” either from the bottom right button, or navigation menu “File > Open Dataset”, you are prompted to open a dataset or an exchange catalog. If you open an exchange catalog (CATALOG.XML) the “Exchange Catalog Browser” will open shown in Figure 11. The program will infer the feature catalog and portrayal catalog from the product specification information and dataset discovery metadata in the exchange catalog.

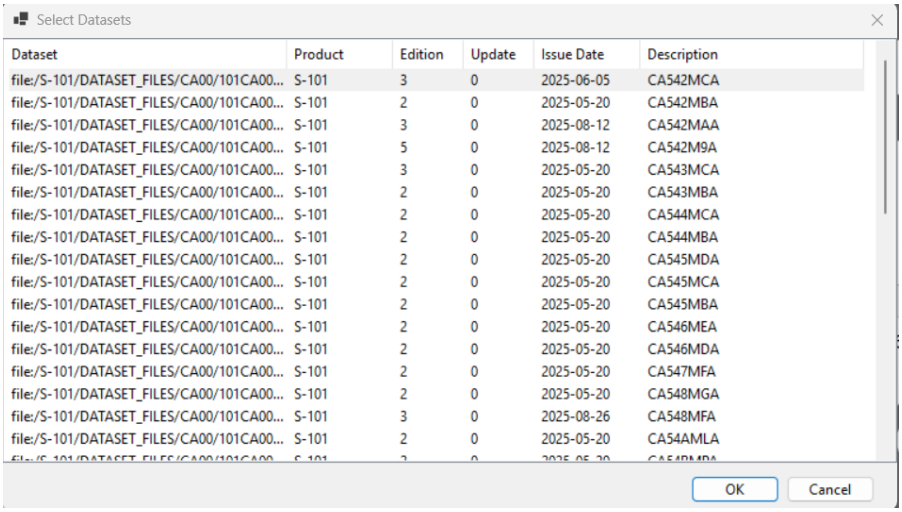


Figure 11. Exchange Catalog Browser

You can change the feature and portrayal catalogues in the "Select Catalogs" dialog shown in Figure 12, or pick a feature and portrayal catalogue from a file on your system by selecting "Browse…" button to open a file explorer if the correct ones are not found from exchange catalogue metadata.

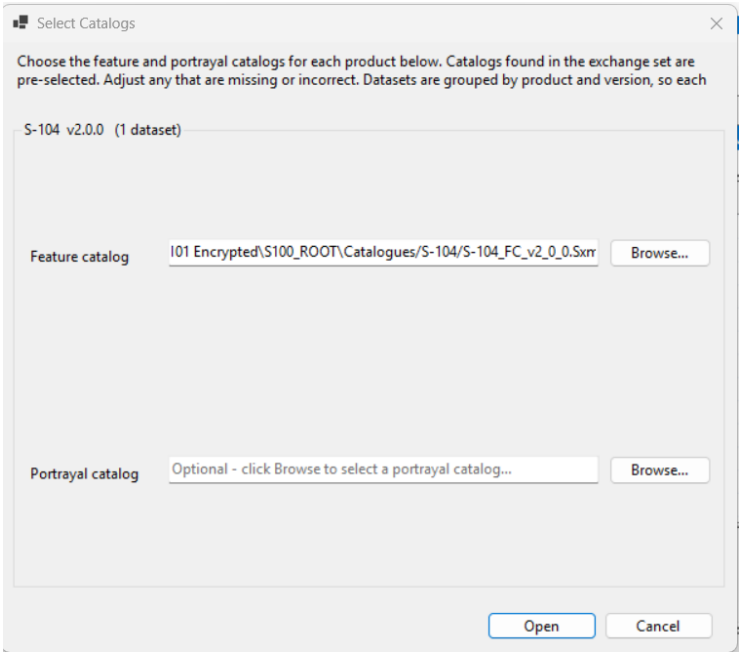


Figure 12. Feature and Portrayal Catalog Browser

The portrayal catalog choice is shown as optional, so as to be able to explore the S-100 feature and dataset record content of products that may not have a portrayal catalog available e.g. S-104 Water Level Information for Surface Navigation.

4.1.1.2.1 ISO-8211 Browse Mode

This mode will display a data set encoding in its raw ISO-8211 format (assumes the data set is encoded in ISO-8211 format). By selecting this mode, an "ISO-8211 Browser" tab is added to the available "Source

Tabs" shown in Figure 2. For further information regarding what is available in this browsing mode, see section 5.1.1. The loading procedure that should be followed for this browse mode is as follows:

1. Select "File → Open Dataset..." from the S-100 Viewer menu.
2. Select the desired S-101 data file from the file browser, then select "Open".
3. In the file browser that prompts for selection of the feature catalogue, select "Cancel".

4.1.1.2.2 S-100 Browse Mode

This browse mode uses a selected feature catalogue to browse an S-100 data set file as a set of S-100 features. By selecting this mode, the ISO-8211 Browse Mode is available and a "S-100 Browser" tab is added to the available "Source Tabs" identified in Figure 2. For further information regarding what is available in this browsing mode, see section 5.1.3. The loading procedure for this browse mode is:

1. Select "File → Open Dataset..." from the S-100 Viewer menu.
2. Select the desired S-100 data file, or an exchange catalog from the file browser, then select "Open".
3. Select the feature catalogue that will be used from the file browser, then select "Open".
4. In the file browser that prompts for selection of the portrayal catalogue, select "Cancel".

4.1.1.2.3 Portrayal Mode

This browse mode uses a selected feature and portrayal catalogue to open and display an S-100 data set file using a graphic portrayal in the Chart Area (See 6.0). By selecting this mode, the ISO-8211 Browse Mode or HDF5 Browse Mode, and S-100 Browse Mode are available and a "Portrayal" tab is added to the available "Source Tabs" identified in Figure 2. For further information regarding what is available in this browsing mode, see section 5.1.4. The loading procedure for this browse mode is:

1. Select "File → Open Dataset..." from the S-100 Viewer menu.
2. Select the desired S-100 data file from the file browser, then select "Open".
3. Select the feature catalogue that will be used from the file browser, then select "Open".
4. Select the portrayal catalogue that will be used from the file browser, then select "Open".

The type of portrayal catalogue selected in step 4 of the above procedure determines whether the portrayal will be generated using XSLT or Lua.

4.1.1.3 Open Feature Catalogue

The S-100 Viewer supports browsing of S-100 feature catalogues. For further information regarding what is available in this browsing mode, see section 5.2. The procedure to open and browse a feature catalogue is:

1. Select "File → Open Feature Catalogue..." from the S-100 Viewer menu.
2. Select the desired S-100 feature catalogue using the file browser.

4.1.1.4 Screenshot

The menu entry "File → Screenshot..." supports capturing an image of the S-100 Viewer application window that can be saved as an image (.jpg) file. A file browser will prompt for a file name and folder location to save the screen image.

4.1.1.5 Exit

To exit the S-100 Viewer, select "File → Exit" from the S-100 Viewer menu, or click the window close button.

4.1.2 View Menu

The View menu provides the ability to change how S-101 data information is displayed on the screen through proper calibration of the monitor. The options that are available in the View Menu are shown in Figure .

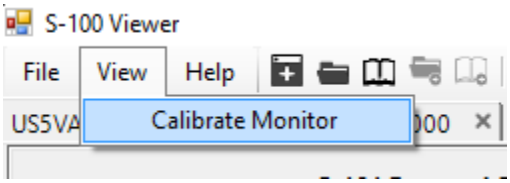


Figure 13. View Menu Options

4.1.2.1 Calibrate Monitor

The monitor must be calibrated for display scale to be indicated correctly, and for symbols and text drawn to the screen to be sized accurately. The calibration lets the program determine how large each pixel on the monitor is, and therefore how many pixels to use when drawing each portrayal item.

Normally the viewer will automatically calibrate the monitor, but in certain instances it may be necessary to manually calibrate the monitor using this function.

4.1.3 Portrayal Menu

The portrayal menu provides the option to enable or disable a Lua portrayal cache. The concept of a Lua portrayal cache is described in S-100 Part 9a, and is primarily intended to speed up Lua portrayal generation when context parameters are changed.

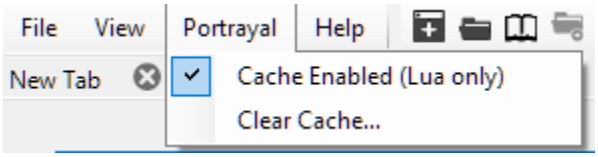


Figure 14. Portrayal Menu Options

These options are currently disabled.

Help Menu

The Help menu provides information about the S-100 Viewer. The options that are available in the Help Menu are shown in **Error! Reference source not found.**15.

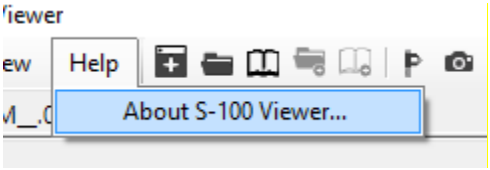


Figure 15. Help Menu Options

4.1.3.1 About S-100 Viewer

The menu entry "Help → About S-100 Viewer..." opens a window that shows S-100 Viewer information.

4.2 Source Toolbar

The Source Toolbar provides much of the same functionality to the user that the Menus provide but with a more concise view. **Error! Reference source not found.**16 provides a detailed image indicating the names of each of the buttons. For ease-of-use, tool tips for these buttons are available by hovering the cursor over the desired button.

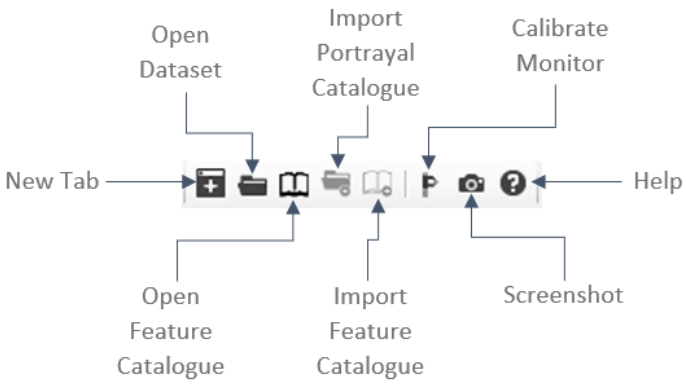


Figure 16. Source Toolbar Buttons

4.2.1 New Tab

See 4.1.1.1.

4.2.2 Open Dataset

See 4.1.1.2.

4.2.3 Open Feature Catalogue

See 4.1.1.3.

4.2.4 Import Portrayal Catalogue

Unimplemented.

4.2.5 Import Feature Catalogue

Unimplemented.

4.2.6 Calibrate Monitor

See 4.1.2.1.

4.2.7 Screenshot

See 4.1.1.4.

4.2.8 Help

See 4.1.3.1.

4.3 Source Tabs

The Source Tabs, as seen in Figure , allow for switching between data set browsing modes within a data set tab. Note: the availability of each of the source tabs is dependent on the items selected during data set loading (See 4.1.1.2).

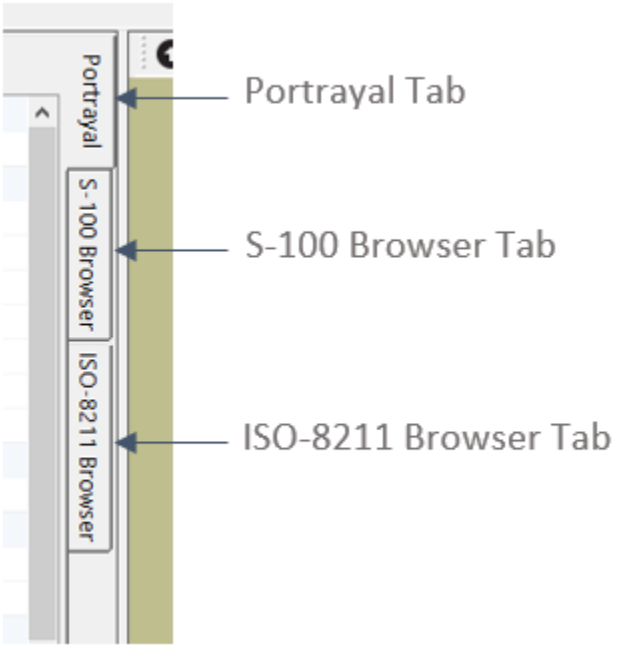


Figure 17. Source Tabs

4.3.1 Portrayal Tab

See 5.1.4.

4.3.2 S-100 Browser Tab

See 5.1.3.

4.3.3 ISO-8211 Tab

See 5.1.1.

4.4 Chart Area Toolbar

The Chart Area Toolbar, as seen in Figure 18, provides buttons, options, and information to interact with the Chart Area (See 6.0).

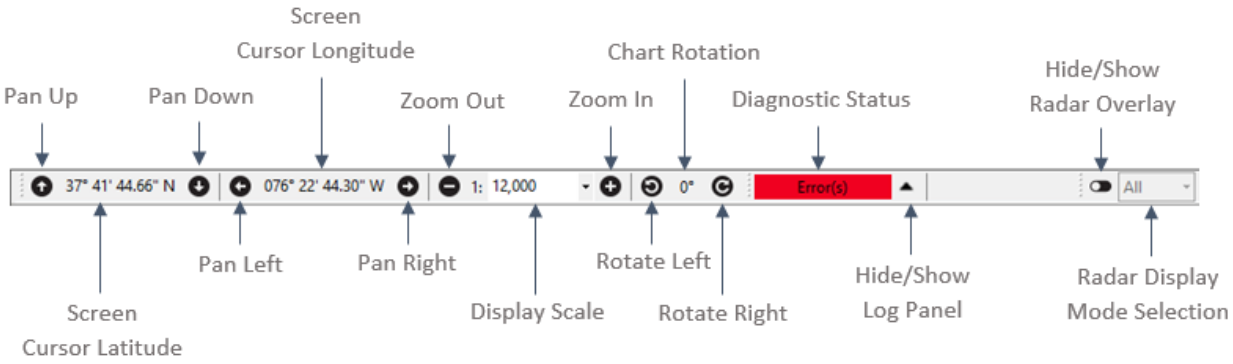


Figure 18. Chart Area Toolbar Controls

4.4.1 Pan Up

This panning operation allows for movement of the image in an up-direction, taking into account any rotation that may previously be applied to the image. In addition to the toolbar button, when the chart area has been selected the user can press the UP arrow key on the keyboard to perform the same action. Each panning operation results in the movement of the image by 50% of the current height.

4.4.2 Pan Down

This panning operation allows for movement of the image in a down-direction, taking into account any rotation that may previously be applied to the image. In addition to the toolbar button, when the chart area has been selected the user can press the DOWN arrow key on the keyboard to perform the same action. Each panning operation results in the movement of the image by 50% of the current height.

4.4.3 Pan Left

This panning operation allows for movement of the image in a left-direction, taking into account any rotation that may previously be applied to the image. In addition to the toolbar button, when the chart

area has been selected the user can press the LEFT arrow key on the keyboard to perform the same action. Each panning operation results in the movement of the image by 50% of the current width.

4.4.4 Pan Right

This panning operation allows for movement of the image in a right-direction, taking into account any rotation that may previously be applied to the image. In addition to the toolbar button, when the chart area has been selected the user can press the RIGHT arrow key on the keyboard to perform the same action. Each panning operation results in the movement of the image by 50% of the current width.

4.4.5 Screen Cursor Latitude

This display represents the current latitude for where the cursor is drawn on the Chart Area (See 6.0), and will automatically update as the cursor is moved. The field may be clicked to change the coordinate format to one of the following formats:

1. Degrees Minutes Seconds (Default)
2. Decimal Degrees
3. Degrees Decimal Minutes

4.4.6 Screen Cursor Longitude

This display represents the current longitude for where the cursor is drawn on the Chart Area (See 6.0), and will automatically update as the cursor is moved. The field may be clicked to change the coordinate format to one of the following formats:

1. Degrees Minutes Seconds (Default)
2. Decimal Degrees
3. Degrees Decimal Minutes

4.4.7 Zoom Out

This zooming operation allows for moving out of a rendered image drawn on the Chart Area (See 6.0). In addition to the toolbar button, when the chart area has been selected the user can press the MINUS (-) key on the keyboard to simply zoom out. Likewise, by holding the CONTROL keyboard key and double-clicking the left mouse button, the resulting image is zoomed out and centered about the cursor location the double-click was performed at. Each zooming operation results in increasing the display scale by a factor of 2.

4.4.8 Zoom In

This zooming operation allows for moving in to a rendered image drawn on the Chart Area (See 6.0). In addition to the toolbar button, when the chart area has been selected the user can press the PLUS (+) key on the keyboard to simply zoom in. Likewise, by holding the CONTROL keyboard key and double-clicking the left mouse button, the resulting image is zoomed in and centered about the cursor location the double-click was performed at. Each zooming operation results in decreasing the display scale by a factor of 2.

4.4.9 Display Scale

Indicates the size of features drawn on the screen relative to their real-world size. The numerator is implied. For example, a display scale of 10,000 indicates a 1:10,000 display scale, where measuring the length of a pier on a screen and multiplying by 10,000 will result in the actual length of the pier. This assumes the monitor has been previously calibrated (See 4.1.2.1).

4.4.10 Rotate Left

This rotate operation allows for a perceived counter-clockwise rotation of a rendered image drawn in the Chart Area (See 6.0). The operation changes the direction of "North" relative to the center of the portrayal in 45 degree increments. In addition to the tool bar button, when the chart area has been selected the user can press and hold the CONTROL key and press the LEFT arrow key.

4.4.11 Rotate Right

This rotate operation allows for a perceived clockwise rotation of a rendered image drawn in the Chart Area (See 6.0). The operation changes the direction of "North" relative to the center of the portrayal in 45 degree increments. In addition to the tool bar button, when the chart area has been selected the user can press and hold the CONTROL key and press the RIGHT arrow key.

4.4.12 Chart Rotation

This display field indicates the direction of "North" relative to the center of the portrayal. For example, 90 indicates that north is to the right. This field will automatically update as rotation is applied to the rendered image or a drop down selection menu is available by clicking the arrow.

4.4.13 Diagnostic Status

This status field indicates if the application encountered warnings or errors when loading the S-100 data set and will display this as seen in Figure . This status field can also be clicked to open and close the Log Panel (See 7.0).

4.4.14 Hide/Show Log Panel

This button can be clicked to open and close the Log Panel (See 7.0).

4.4.15 Hide/Show Radar Overlay

This button is used to hide or show the radar overlay as seen in Figure 19. The overlay is in a fixed location in the visible Chart Area and shows decreasing transparency values in the clockwise direction. As transparency decreases, items in the Below Radar display mode (See 4.4.16) become less visible whereas items in the Above Radar display mode remain visible.

This option is intended to support verification of display plane assignment for drawing instructions generated by a portrayal catalogue.

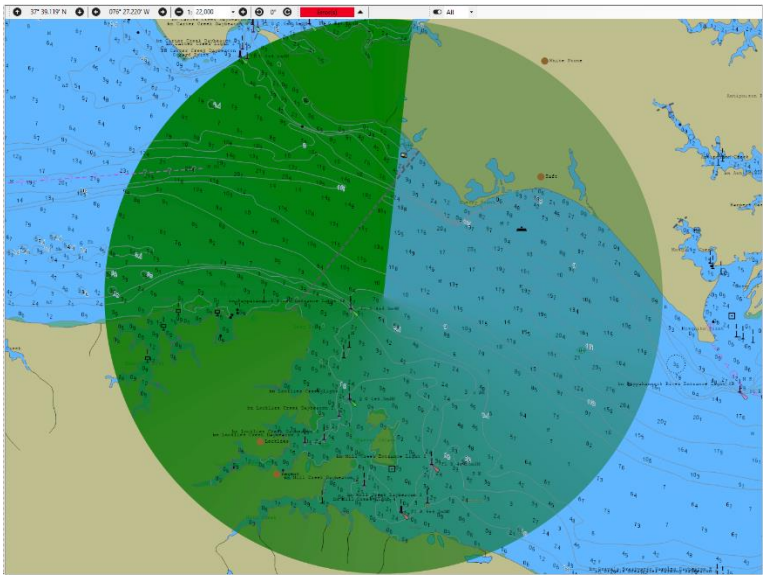


Figure 19. Radar Overlay Enabled

4.4.16 Radar Display Mode Selection

The selection filters drawing instructions by the selected display plane(s). It can be used to help verify display plane assignment for drawing instructions generated by a portrayal catalogue.

4.4.16.1 All

This radar display mode is the default Radar Display Mode selection and portrays all drawing instructions regardless of display plane assignment. Figure 20 shows the portrayal using the “All” radar display mode.

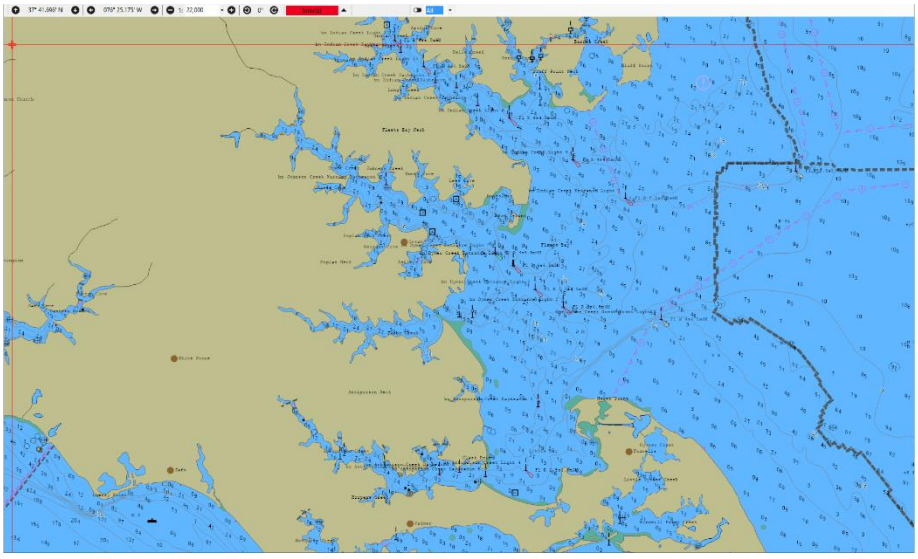


Figure 20. "All" Radar Display Mode Selection

4.4.16.2 Above

This radar display mode portrays all items that have been included in a display plane which is above the Radar image. Drawing instructions assigned to display planes below the Radar image will not be displayed. Figure 21 shows the portrayal using the “Above” radar display mode.

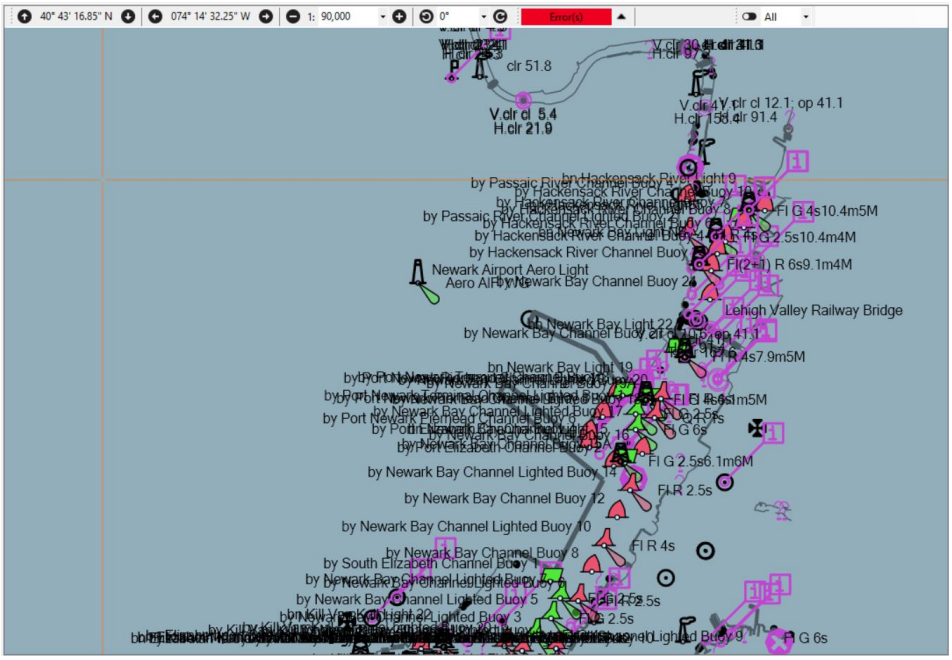


Figure 21. "Above" Radar Display Mode Selection

4.4.16.3 Below

This radar display mode portrays all items that have been included in a display plane which is below the Radar image. Drawing instructions assigned to display planes above the Radar image will not be displayed. Figure 22 shows the portrayal using the “Below” radar display mode.

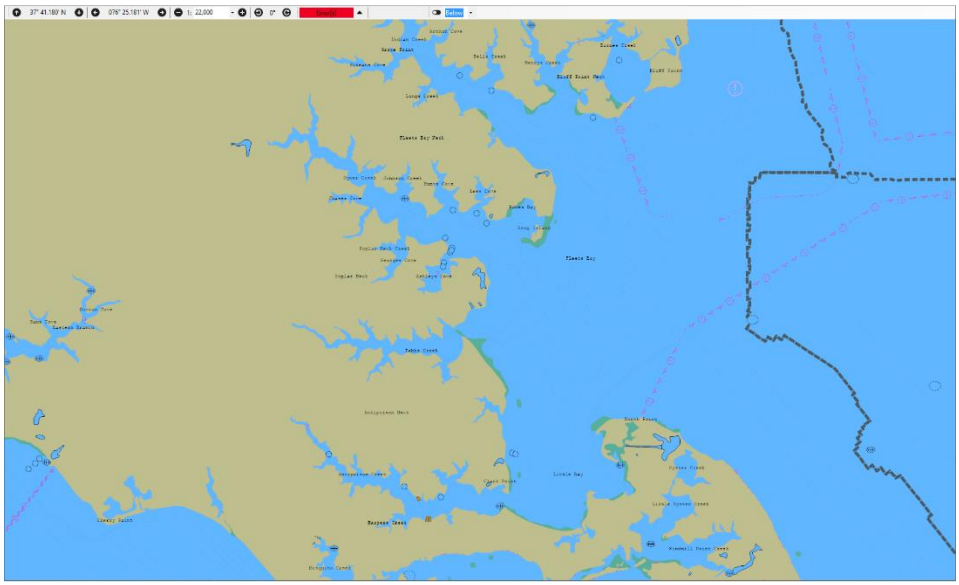


Figure 22. "Below" Radar Display Mode Selection

5.0 PRESENTATION BROWSERS

5.0 PRESENTATION BROWSERS

The S-100 Viewer supports multiple browsing views each with their own unique presentations. The browser that is presented depends on the file types that were loaded into the S-100 Viewer. The supported presentation browsers include the ISO-8211, S-100, Portrayal, and Feature Catalogue Browsers.

5.1 Data Presentation Browsers

The S-100 Viewer supports multiple browsing views for any given S-100 data set file, though currently limited to S-101 data set files. The browsers that are available to the user are determined by whether a feature or portrayal catalogues were selected during the loading procedure (See 4.1.1.2). When proper portrayal and feature catalogues are loaded, the data presentation tabs that are available include the ISO-8211 Browser, S-100 Browser, and Portrayal.

5.1.1 ISO-8211 Browser Tab

The ISO-8211 Browser Tab is available by default when an ISO-8211 encoded S-100 data set file is selected during the data loading procedure (See 4.1.1.2). The browser tab is organized into a three block hierarchical layout as depicted in Figure .

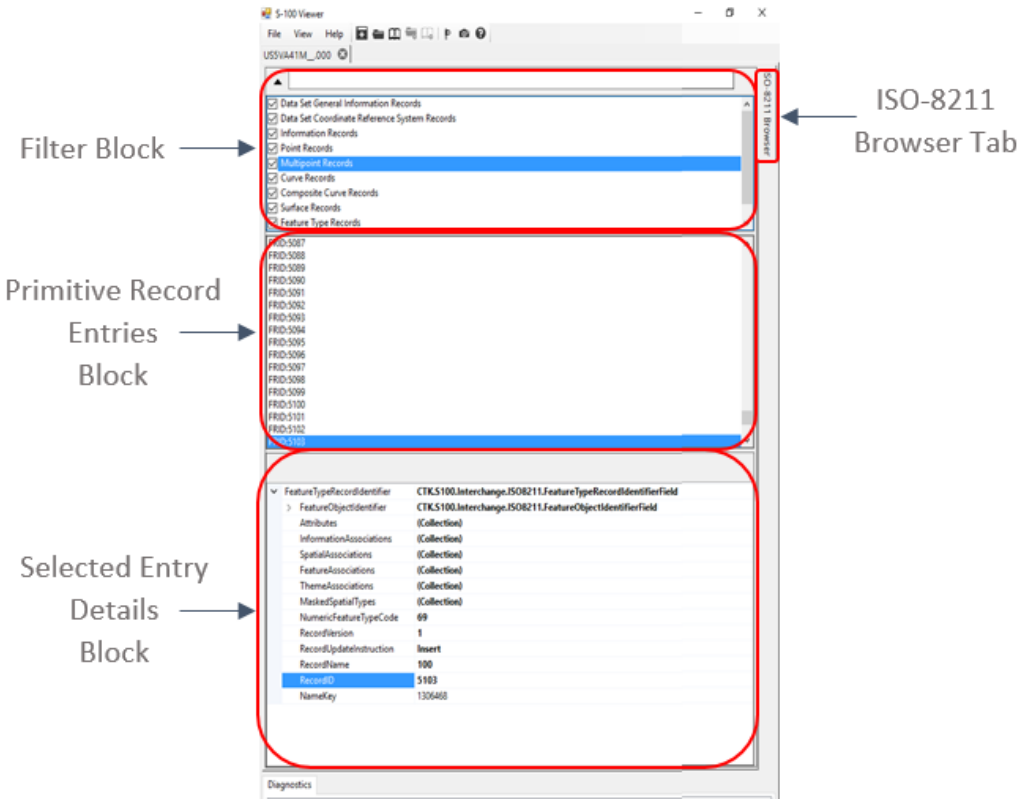


Figure 23. ISO-8211 Browser Tab Layout

5.1.1.1 Filter Block

The filter block allows for the ability to search for matching record information within the loaded ISO-8211 data set. There is a drop down menu that becomes visible by clicking the drop down chevron (See

Figure 24) which defines available record types to filter the data set by. If a filter item is enabled, entries of that record type are populated in the "Primitive Record Entries Block" identified in Figure . By default, all options are enabled.

When one or more options in the list are disabled, then a funnel icon will be displayed to the right of the filter text field (See Figure 24) and the searchable options in the "Primitive Record Entries Block" will be narrowed down accordingly.

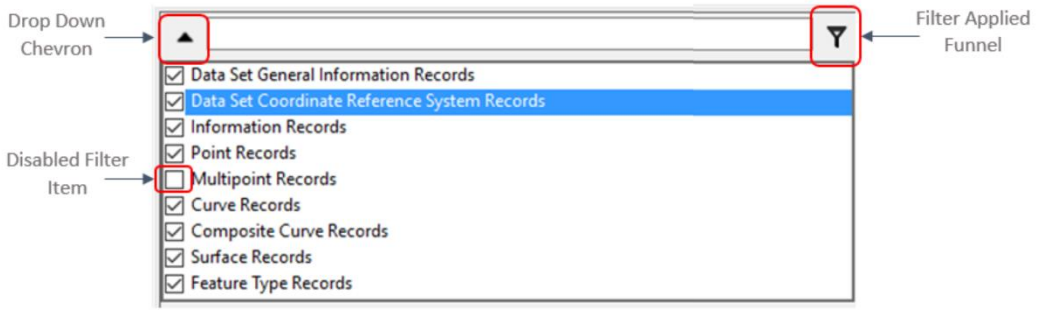


Figure 24. Filter Item Disabled

The search algorithm allows for multiple search terms in the filter text field to further filter the results displayed in the "Primitive Record Entries Block". The algorithm will AND each of the inputted search terms together to provide a list of specific instances where all terms apply and comply with the search filter types that are enabled in the drop down menu. All record entries that match the filter criteria are displayed as results in the "Primitive Record Entries Block" (See Figure).

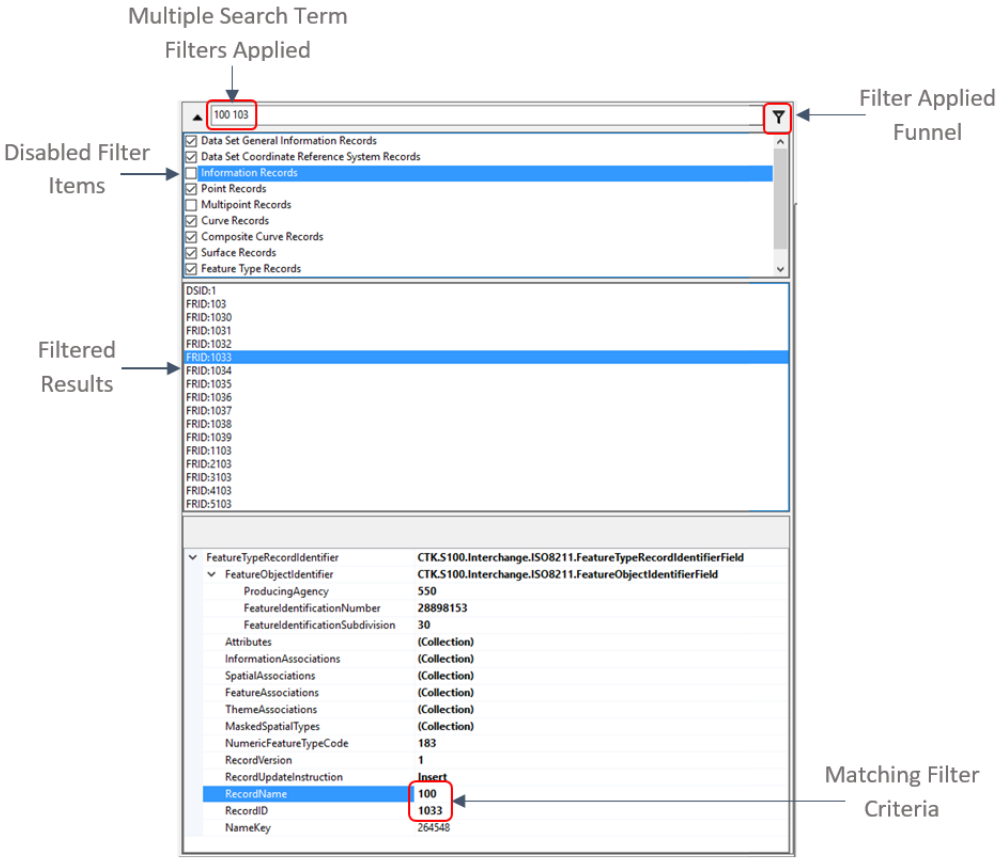


Figure 25. Multiple Filters Applied

5.1.1.2 Primitive Record Entries Block

The “Primitive Record Entries Block” shows all records that can be reviewed at a given time. By default, with no filter criteria entered in the “Filter Block”, then all defined records from the S-100 data set are listed.

5.1.1.3 Selected Entry Details Block

The “Selected Entry Details Block” shows the details for the selected data set entry as seen in the data set file. Metadata, associations, and attributes of the selected item entry are all displayed as shown in Figure .

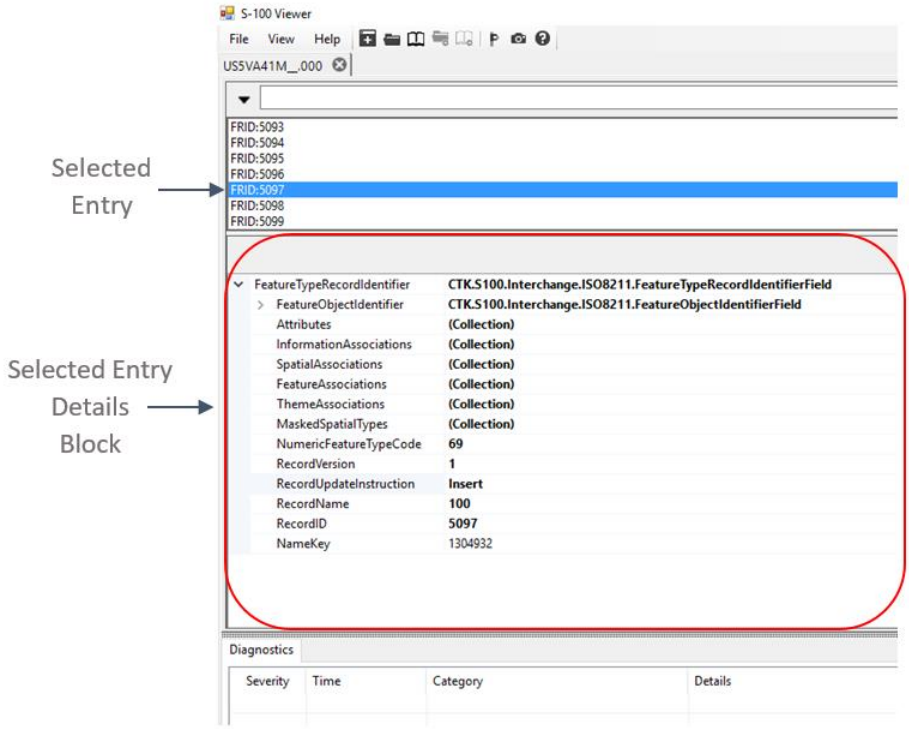


Figure 26. Selected Entry Details Block

5.1.1.3.1 Selected Entry Details Block Controls

Included in the “Selected Entry Details Block” are controls used to further expand and view nested entry details. The available controls have been identified below in Figure .

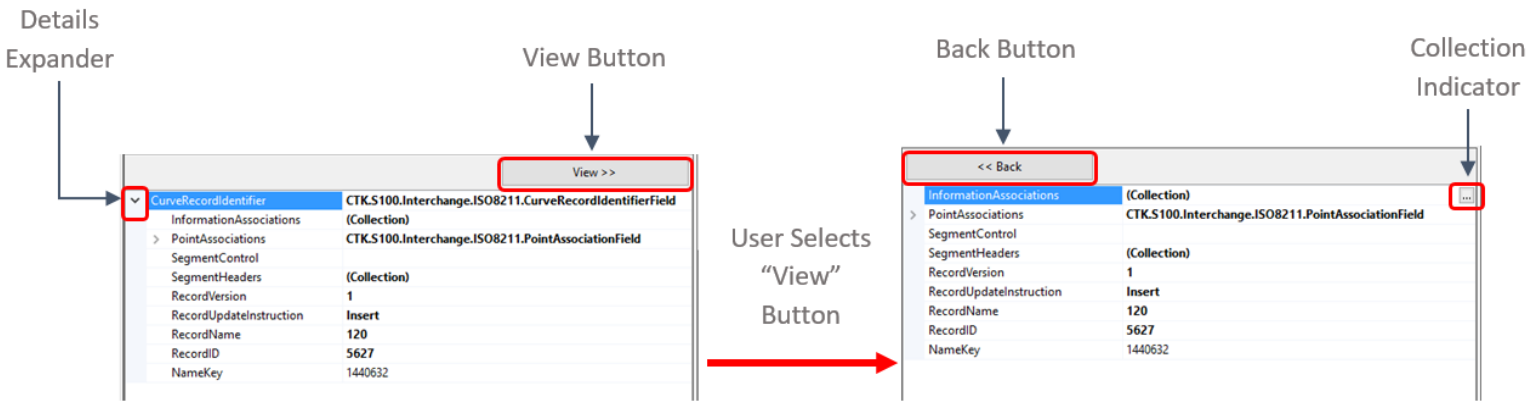


Figure 27. Selected Entry Details Block Controls

The controls that are available and their corresponding uses are as follows:

1. Details Expander – Expands nested details relating to the currently selected item.

2. View Button – Expands nested details relating to the currently selected item in another window. This control is only available if the selected detail contains nested details.

3. Back Button – Used to return to the previous window before the expansion operation occurred. This control is only available if the user has used the “View Button” to expand an items details. This concept is indicated in Figure .

4. Collection Indicator – Spawns a collection dialog to view the contents of a collection. This control is only available if an entry detail is a collection of objects. The collection dialog can be seen below in Figure .

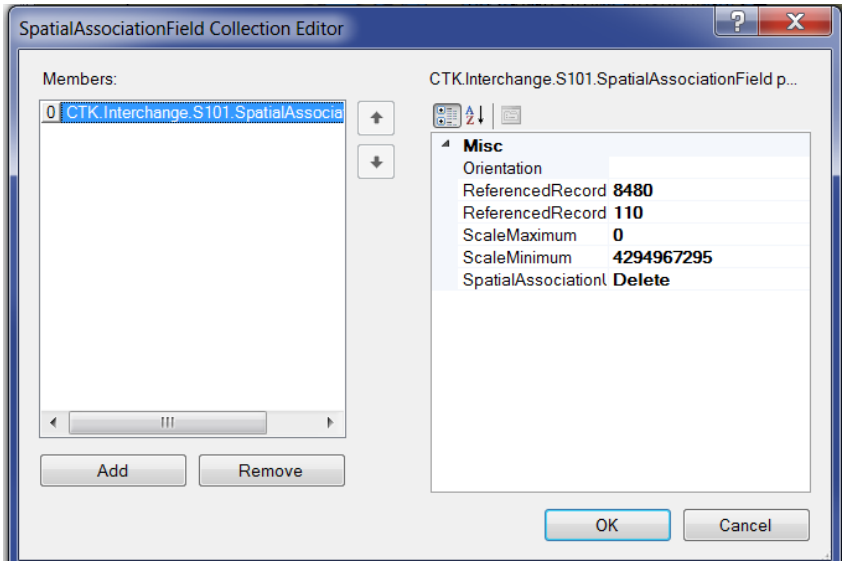


Figure 28. ISO-8211 Collection Dialog

5.1.2 HDF5 Browser Tab

The HDF5 Browser Tab is available by default when an HDF5 encoded S-100 data set file is selected during the data loading procedure (See 4.1.1.2).

The HDF5 Browser tab is split into 2 sections for viewing the HDF5 elements and their attributes.

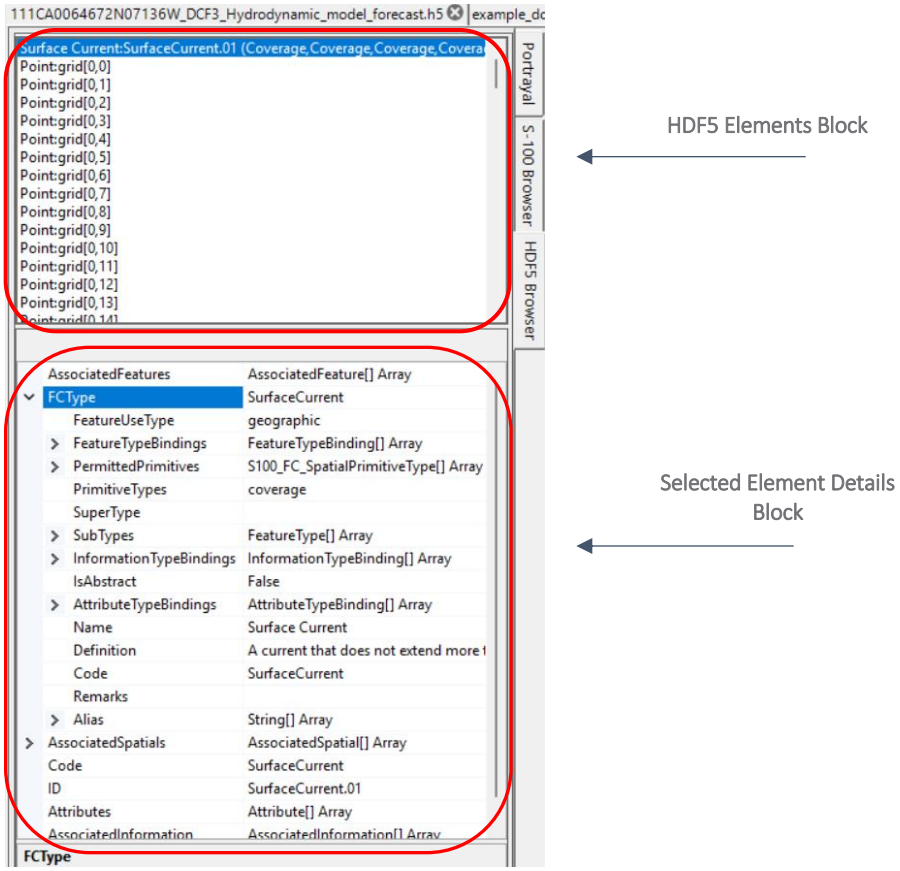


Figure 29. HDF5 Browser Tab

5.1.2.1 HDF5 Elements Block

The "HDF5 Elements Block" shows all elements contributing to the HDF5 data model from a data set file. There is no text filter block for the HDF5 elements, but the listing will respond by filtering elements out when a geographic area is highlighted on the chart with spatial coordinates referenced by the highlighted area.

5.1.2.2 Selected Entry Details Block

The "Selected Entry Details Block" shows the details for the selected data set entry as seen in the data set file. Metadata, associations, and attributes of the selected item entry are all displayed as shown in Figure 29.

5.1.3 S-100 Browser Tab

The S-100 Browser tab provides an area for reviewing feature records loaded from a S-100 data set file, though currently limited to browsing S-101 data sets. The S-100 Browser Tab is only available when a valid feature catalogue has been successfully selected and loaded (See 4.1.1.2). The browser is organized in a three block hierarchical layout as depicted in Figure 30.

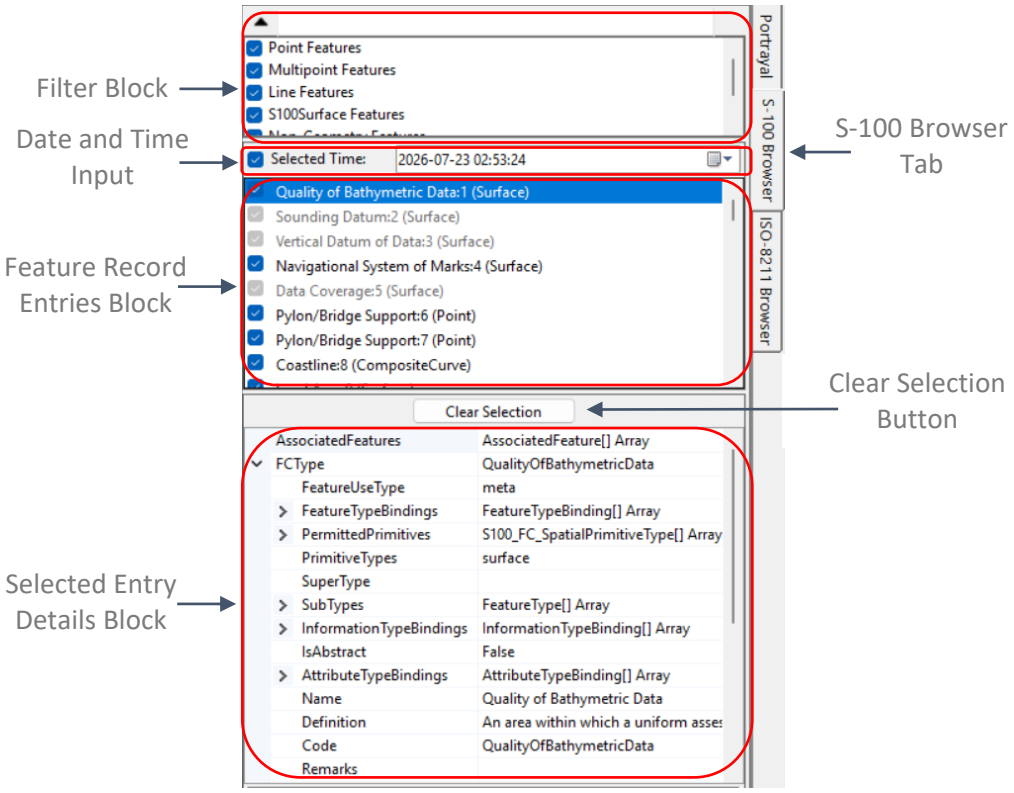


Figure 30. S-100 Browser Tab Layout

5.1.3.1 Filter Block

The filter block allows for the ability to search for matching feature information within the loaded S-100 data set. There is a drop down menu that becomes visible by clicking the drop down chevron (See Figure 31) which defines available feature types to filter the data set by. If a filter item is enabled, entries of that feature type are populated in the "Feature Record Entries Block" identified in Figure . By default, all options are enabled.

When one or more options in the list are disabled, then a funnel icon will be displayed to the right of the filter text field (See Figure 31) and the searchable options in the "Feature Record Entries Block" will be narrowed down accordingly.

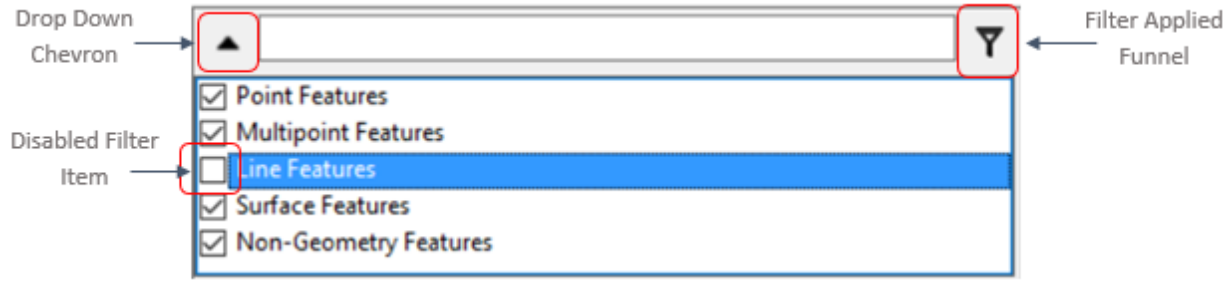


Figure 31. Filter Item Disabled

The search algorithm allows for multiple search terms in the filter text field to further filter the results displayed in the "Feature Record Entries Block". The algorithm will AND each of the inputted search terms together to provide a list of specific instances where all terms apply and comply with the search filter types that are enabled in the drop down menu. All record entries that match the filter criteria are displayed as results in the "Feature Record Entries Block" (See

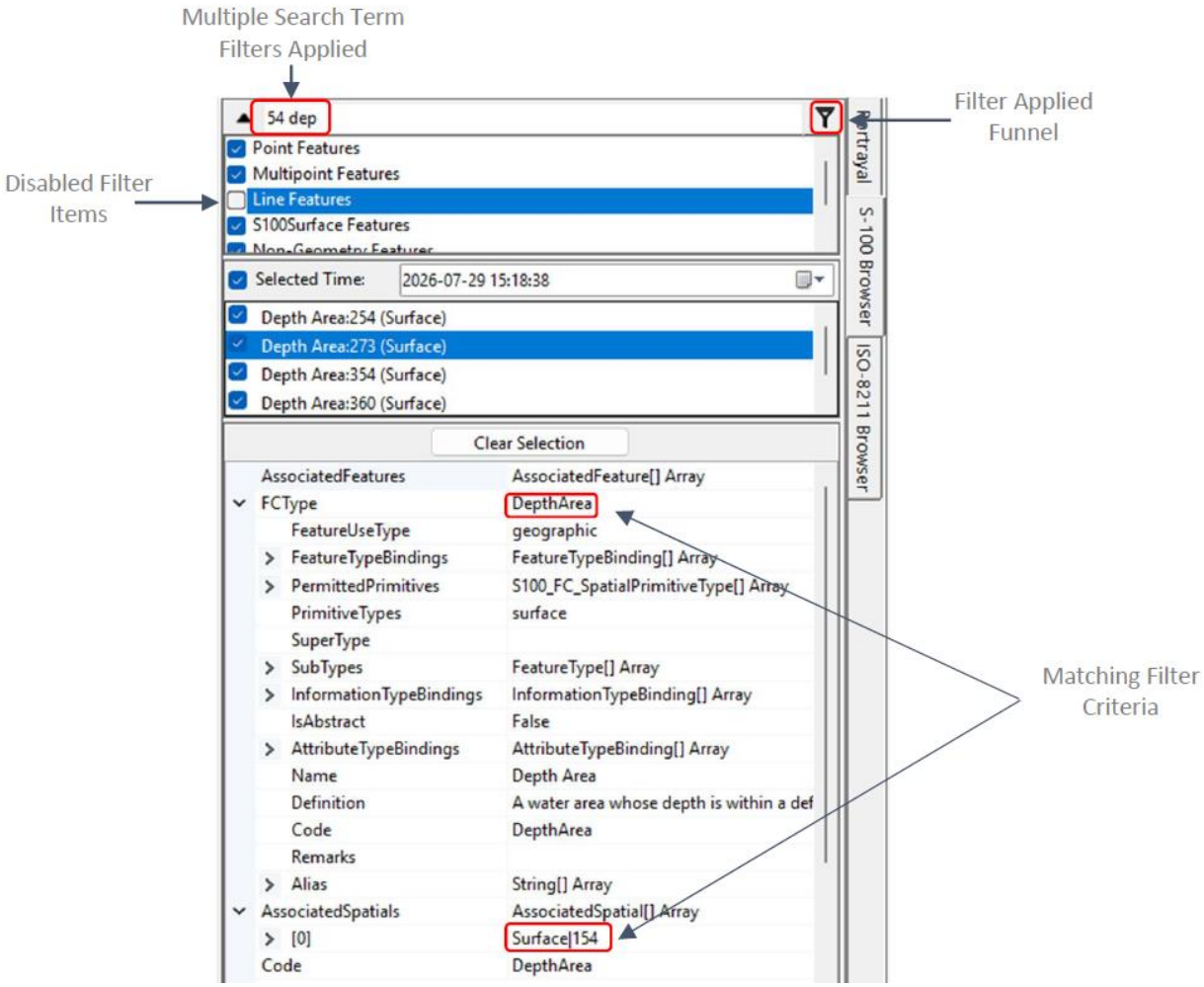


Figure 30).

The filter block also includes a calendar and time picker when GML or ISO-8211 datasets are loaded. When the entire dataset has a valid range, it is restricted to only show dates within that valid range. If a new date and time is picked, the date-dependent features that are not valid for that date and time will drop off the "Feature Record Entries Block" and will be absent from the portrayal view (See Figure 30).

When an HDF5 data set has loaded, the calendar and time picker will be replaced by a list of all applicable coverage dates for a dataset (See Figure 32). If the data set is always valid, there will be no "Time Steps" showing. If an HDF5 data set consists of multiple Z-Index levels for its coverages, the "Z Level" picker will show above the "Feature Record Entries Block". Choose a relevant Z-level, and the values for the "Point:grid" HDF5 Elements will change in the portrayal view and in the HDF5 Browser Attributes for the coverage's grid points.

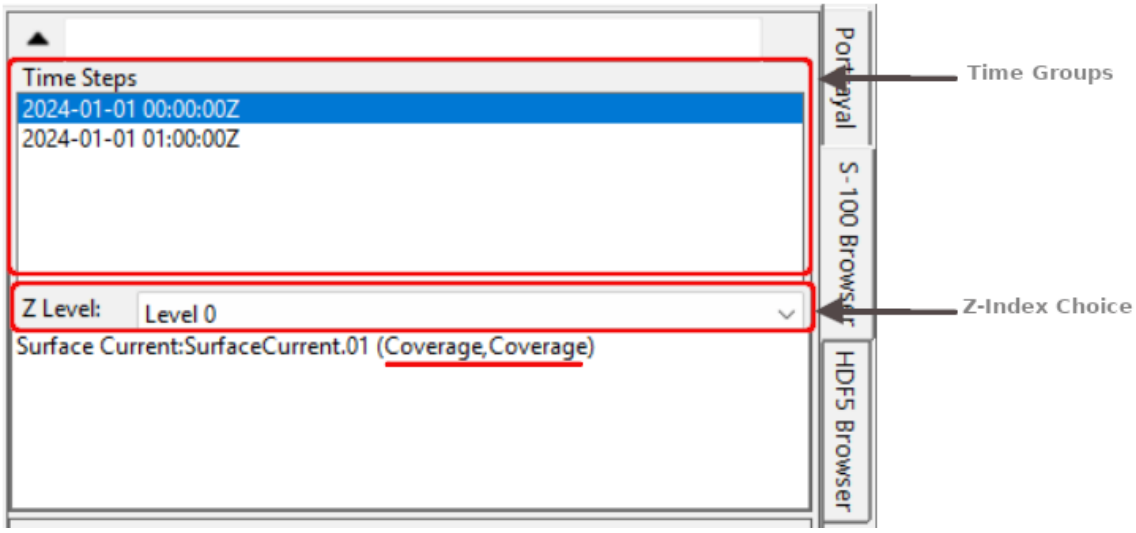


Figure 32. HDF5 Groupings

The Underlined "(Coverage, Coverage)" is showing that there are two spatial associations for the feature, and each one is an independent "Time Group" from the HDF5 data set file.

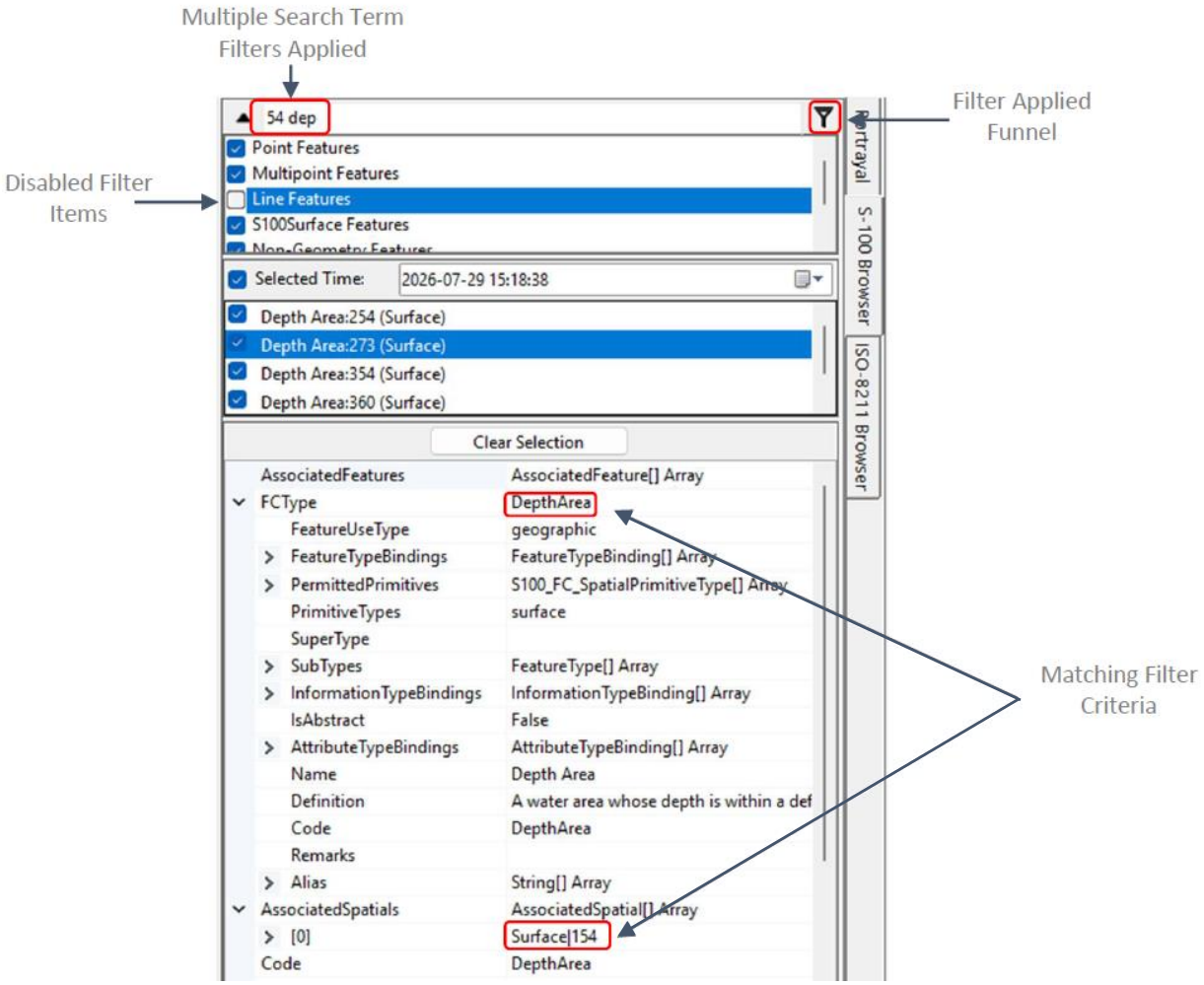


Figure 33. Multiple Filters Applied

5.1.3.2 Feature Record Entries Block

The "Feature Record Entries Block" shows all feature records that can be reviewed at a given time. By default, with no filter criteria entered in the "Filter Block", then all defined feature records from the S-100 data set are listed.

If a portrayal catalogue was selected during the data set opening procedure (See 4.1.1.2), then this area is also used to regenerate the portrayal by moving the center of the portrayal as individual feature records are selected. Also, if a portrayal catalogue has been loaded and an entry has been selected, the "Clear Selection" button identified in Figure will be available and can be used to un-select the current entry selection. This action will clear the "Selected Entry Details Block" and clear any highlights from the charting area (See 6.0).

For **ISO-8211** and **GML** datasets, every feature in the "S-100 Browser" list has a checkbox at the left of its row. A checked box means the feature is drawn on the chart. Unchecking the box by clicking it or selecting with arrow keys and pressing the spacebar will remove the symbol from the chart portrayal. Clicking the feature's name selects the feature by highlighting it in the chart and showing details in the "Selected Details Entry Block" (See Figure 30).

For ISO-8211 and GML datasets, the browser includes a “Selected Time” checkbox next to a date-and-time picker:

- When “Selected Time” checkbox is checked, the chart is drawn as it would appear at the chosen date and time. Features and symbology whose temporal validity does not include that time are not shown. Adjust the date and time with the picker to see how the chart changes.
- When “Selected Time” checkbox is cleared, time is ignored. The date-and-time picker is disabled and every feature that is switched on is drawn regardless of any time information attached to it.

The “Feature Record Entries Block” will only contain features visible given the portrayal settings and context parameters set in the “Portrayal Tab” (See 5.1.4). When “Selected Time” checkbox is checked, features and their checkboxes will be greyed-out and unchangeable if they are not valid for the selected time in the “Date and Time Input” (See Figure 30). To find the greyed-out feature’s portrayal, either search for the time referenced in its attributes (See Figure 34) and set that as the “Selected Time” date-and-time input, or uncheck “Selected Time” checkbox.

5.1.3.3 Selected Entry Details Block

The “Selected Entry Details Block” shows the details for the feature record that has been selected in the “Feature Record Entries Block”. Metadata, associations, and attributes of the selected item entry are all displayed as shown in Figure .

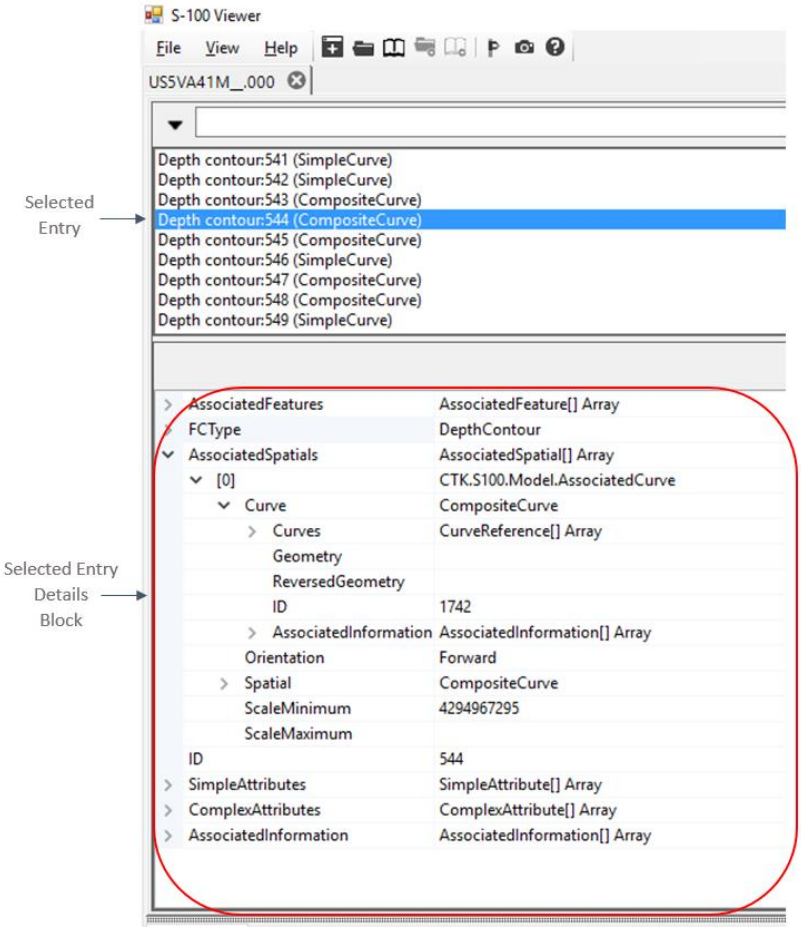


Figure 34. Selected Entry Details Block

5.1.4 Portrayal Tab

The Portrayal Tab provides an area for review and modification of portrayal settings utilized to alter the way the portrayal is drawn to the Chart Area (See 6.0). The tab is divided into six portrayal setting groupings consisting of color palettes, context parameters, display modes, display planes, viewing group layers, and viewing groups. The Portrayal Tab is only available when valid portrayal and feature catalogues have been successfully selected and loaded (See 4.1.1.2).

By clicking a portrayal setting, associated use information is populated in the information panel (See Figure) detailing the changes the particular setting has on the portrayal. Portrayal setting values can be changed by either typing values into the text field or by clicking the drop down arrow located to the right of the display setting value seen in Figure . Changes to any portrayal settings results in an automatic re-draw of the visible Chart Area (See 6.0).

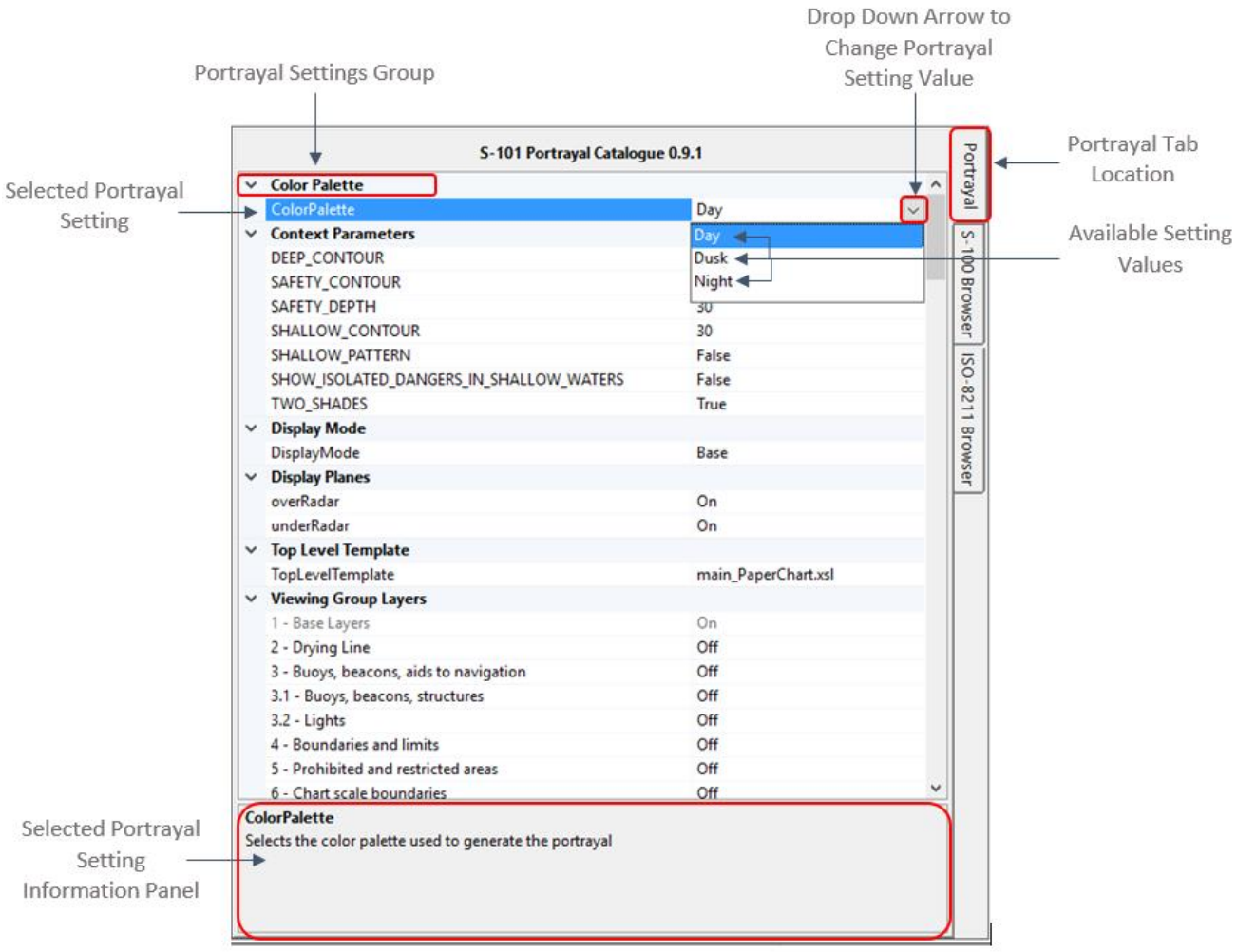


Figure 35. Portrayal Tab Layout

5.1.4.1 Color Palette

The Color Palette portrayal settings group contains a “ColorPalette” portrayal setting that is used to select the color palette used when generating the portrayal. It is important to note that when changing the “ColorPalette” value, there currently isn’t support for changing symbol colors in the portrayal chart area (See 6.0) so symbol colors will remain constant regardless of which color palette value is selected.

5.1.4.2 Context Parameters

The Context Parameters portrayal settings group contains various portrayal settings representing defined options that are retrieved from the loaded portrayal catalogue (See 4.1.1.2). Default values for each of the display settings are indicated in the loaded portrayal catalogue.

5.1.4.3 Display Mode

The Display Mode portrayal settings group contains a “DisplayMode” portrayal setting where the selected value defines the collection of viewing group layers that are visible in the charting area (See 6.0). The

values for this setting are pre-defined in the portrayal catalogue that was loaded (See 4.1.1.2) with the exception of the "Custom" value. The "Custom" value can be manually selected to turn all available Viewing Group Layers and Viewing Groups to visible or the DisplayMode setting will automatically be changed to this value if any of the Viewing Group Layers or Viewing Groups have been altered to where the combination of visible Viewing Groups no longer matches any of the pre-defined DisplayMode values. A visual understanding for how this works can be found in Figure .

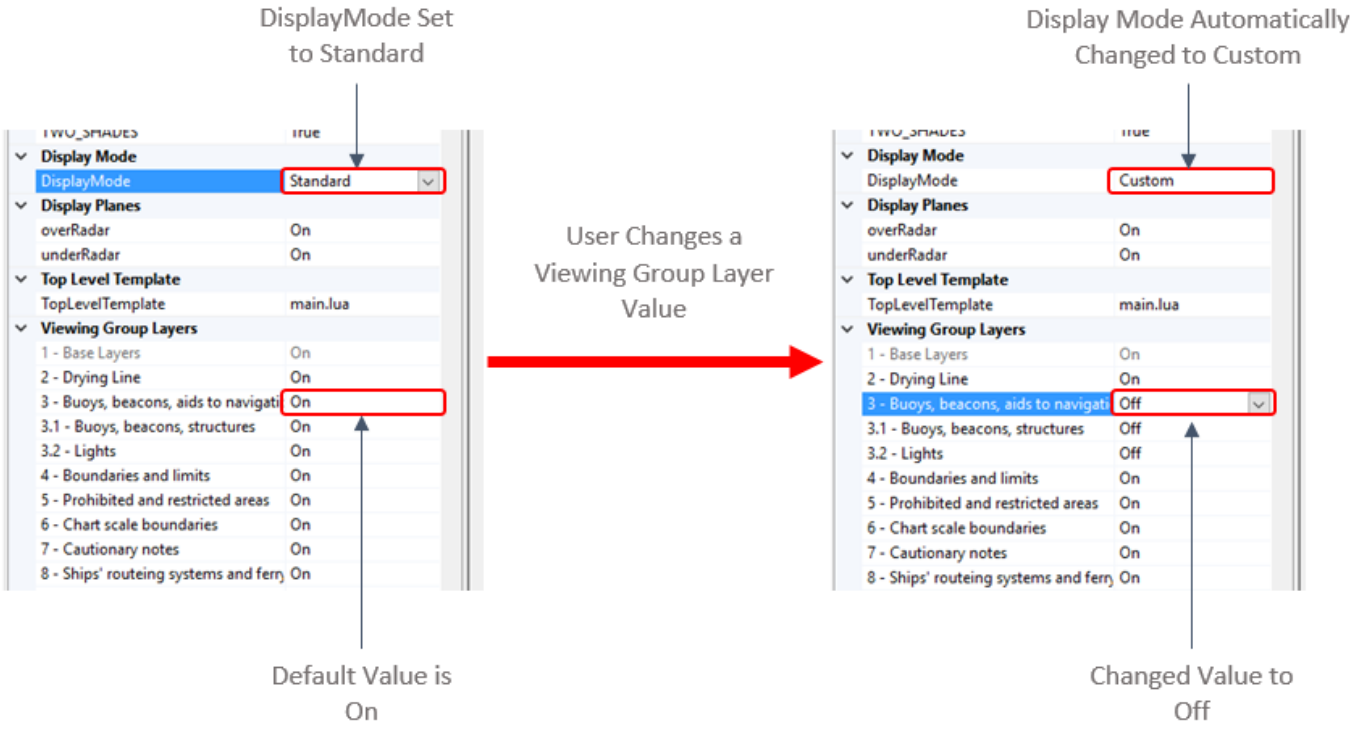


Figure 36. Custom Display Mode Automatically Changing

5.1.4.4 Display Planes

The Display Planes portrayal settings group contains various portrayal settings where each setting represents a high-level selection of Viewing Group Layers (5.1.4.6). The Viewing Group Layers that are associated with each display plane are identified in the loaded portrayal catalogue.

5.1.4.5 Top Level Template

The Top Level Template portrayal settings group contains the "TopLevelTemplate" element where the selected value defines the rule file used to generate drawing instructions. This value is pre-populated with value/s identified in the portrayal catalogue that was loaded (See 4.1.1.2).

5.1.4.6 Viewing Group Layers

The Viewing Group Layers portrayal settings group contains various portrayal settings where each setting represents a different viewing group layer composed of one or more Viewing Groups (See 5.1.4.7). A viewing group layer can be either "On" or "Off" identifying the visibility of the layer as seen in the Chart Area (See 6.0).

Visibility changes to any of the viewing group layers results in an automatic update of the visibility values for the associated Viewing Groups (See 5.1.4.7). Likewise, the "DisplayMode" portrayal setting

(See 5.1.4.3) will be updated to reflect whether the combination of viewing group layers' visibility matches any of the pre-defined display modes. A visual representation of this concept can be seen in Figure .

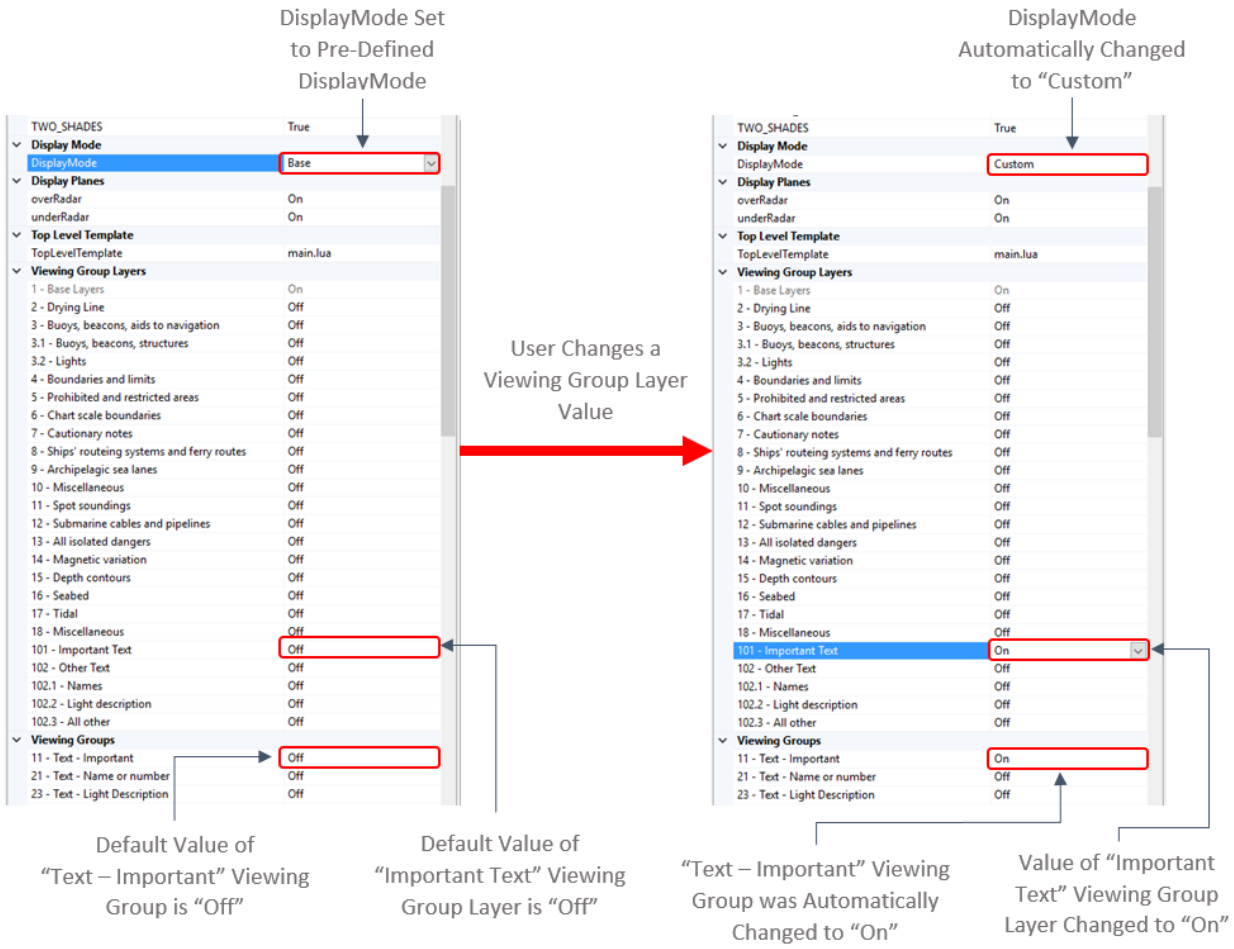


Figure 37. Example of Viewing Group Layer Change

5.1.4.7 Viewing Groups

The Viewing Groups portrayal settings group contains portrayal settings where each setting represents a different viewing group. A viewing group can be either "On" or "Off" identifying the visibility of the viewing group as seen in the Chart Area (See 6.0).

Visibility changes to any of the viewing groups results in an automatic update of the visibility value for the corresponding Viewing Group Layer (See 5.1.4.6) the viewing group is part of. Likewise, the "DisplayMode" portrayal setting (See 5.1.4.3) will be updated to reflect whether the combination of viewing group layers' visibility matches any of the pre-defined display modes. A visual representation of this concept can be seen in Figure .

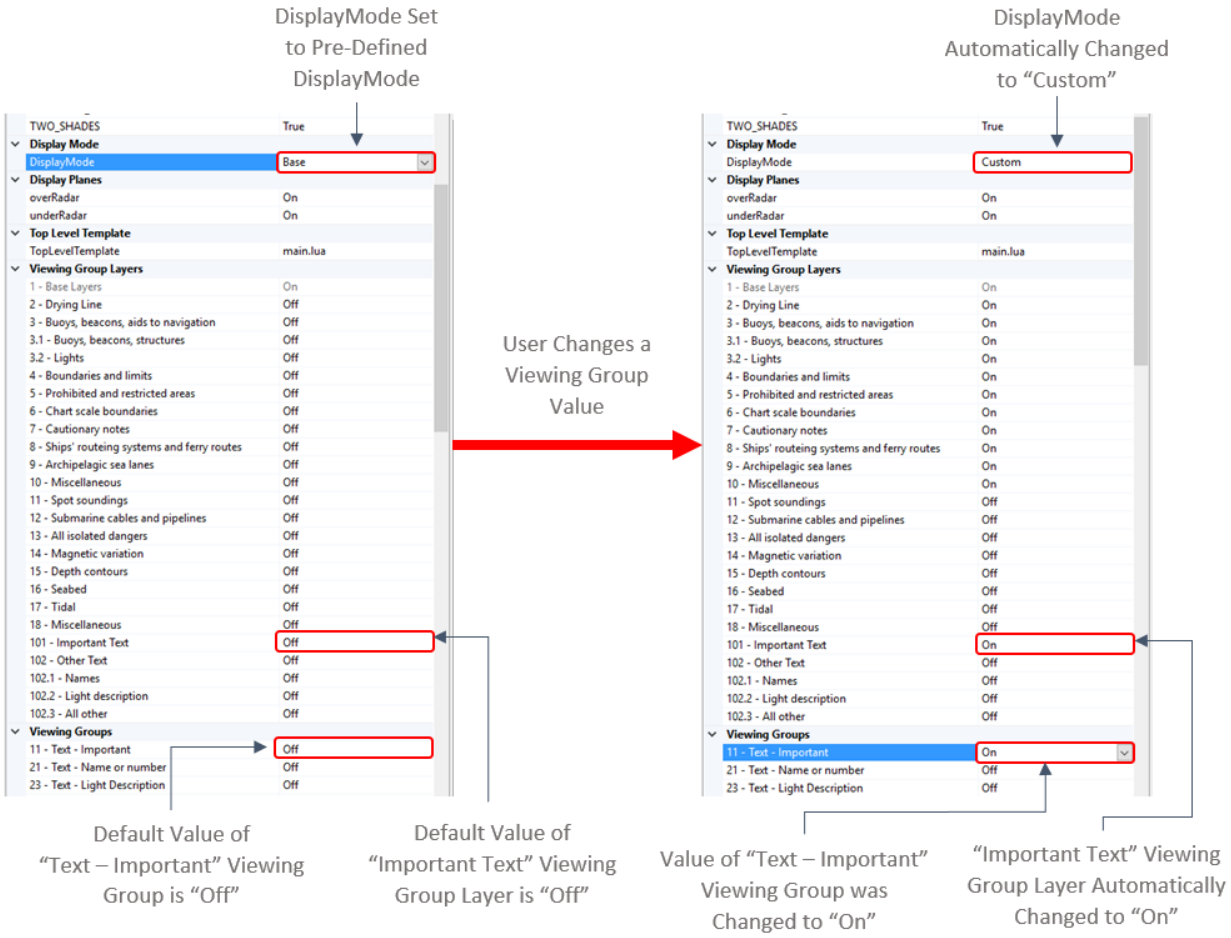


Figure 6. Example of Viewing Group Change

5.2 Feature Catalogue Browser

The Feature Catalogue Browser provides an area for browsing valid S-100 based Feature Catalogues. The browser is organized in a four block hierarchical layout as depicted in Figure .

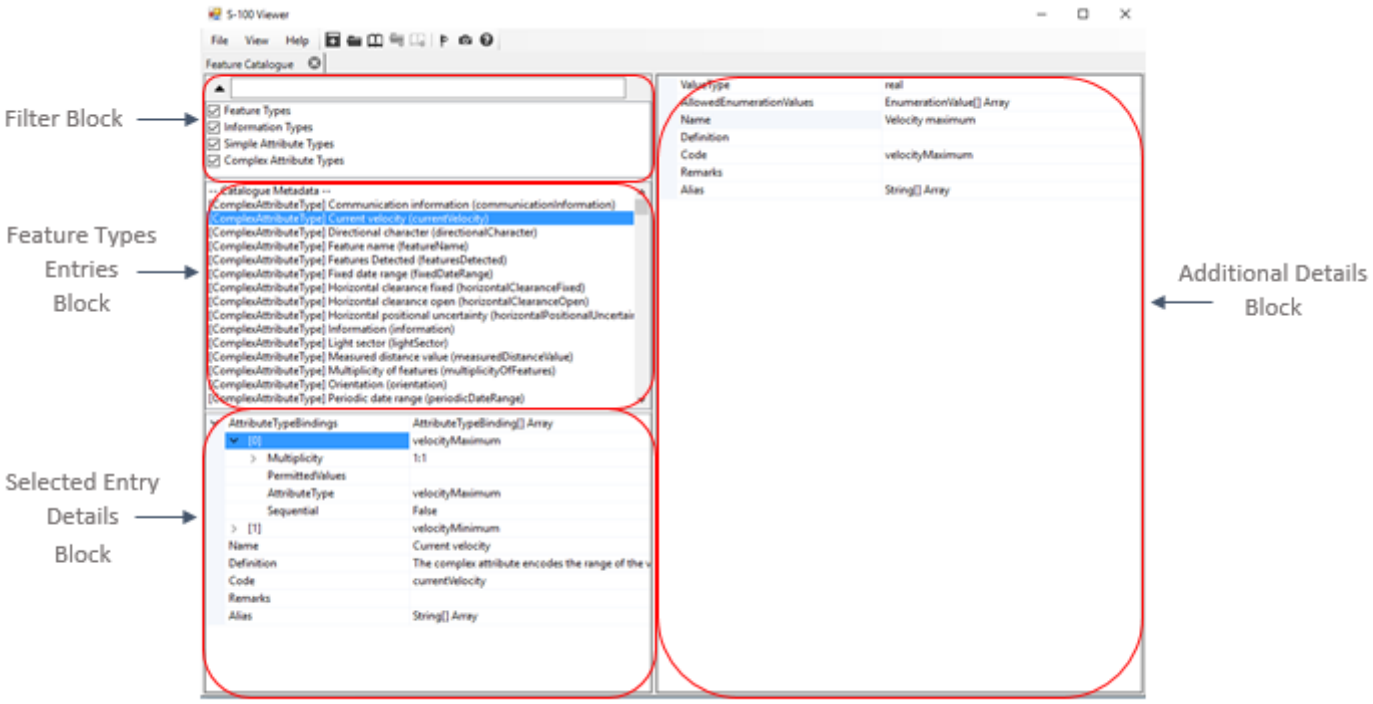


Figure 39. Feature Catalogue Browser

5.2.1 Filter Block

The Filter Block allows for the ability to search for matching feature type information within the loaded S-100 data set. There is a drop down menu that becomes visible by clicking the drop down chevron (See Figure) which defines available types to filter the data set by. If a filter item is enabled, entries of that type are populated in the "Feature Types Entries Block" identified in Figure . By default, all options are enabled.

When one or more options in the list are disabled, then a funnel icon will be displayed to the right of the filter text field (See Figure) and the searchable options in the "Feature Types Entries Block" will be narrowed down accordingly.

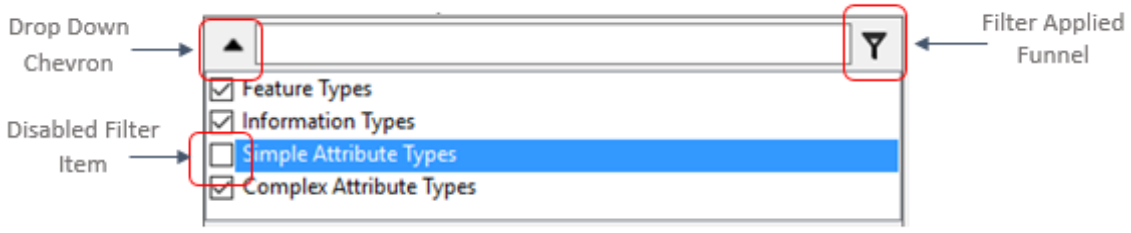


Figure 40. Filter Item Disabled

The search algorithm allows for multiple search terms in the filter text field to further filter the results displayed in the "Feature Types Entries Block". The algorithm will AND each of the inputted search terms together to provide a list of specific instances where all terms apply and comply with the search filter types that are enabled in the drop down menu. All feature type entries that match the filter criteria are displayed as results in the "Feature Types Entries Block" (See Figure 41).

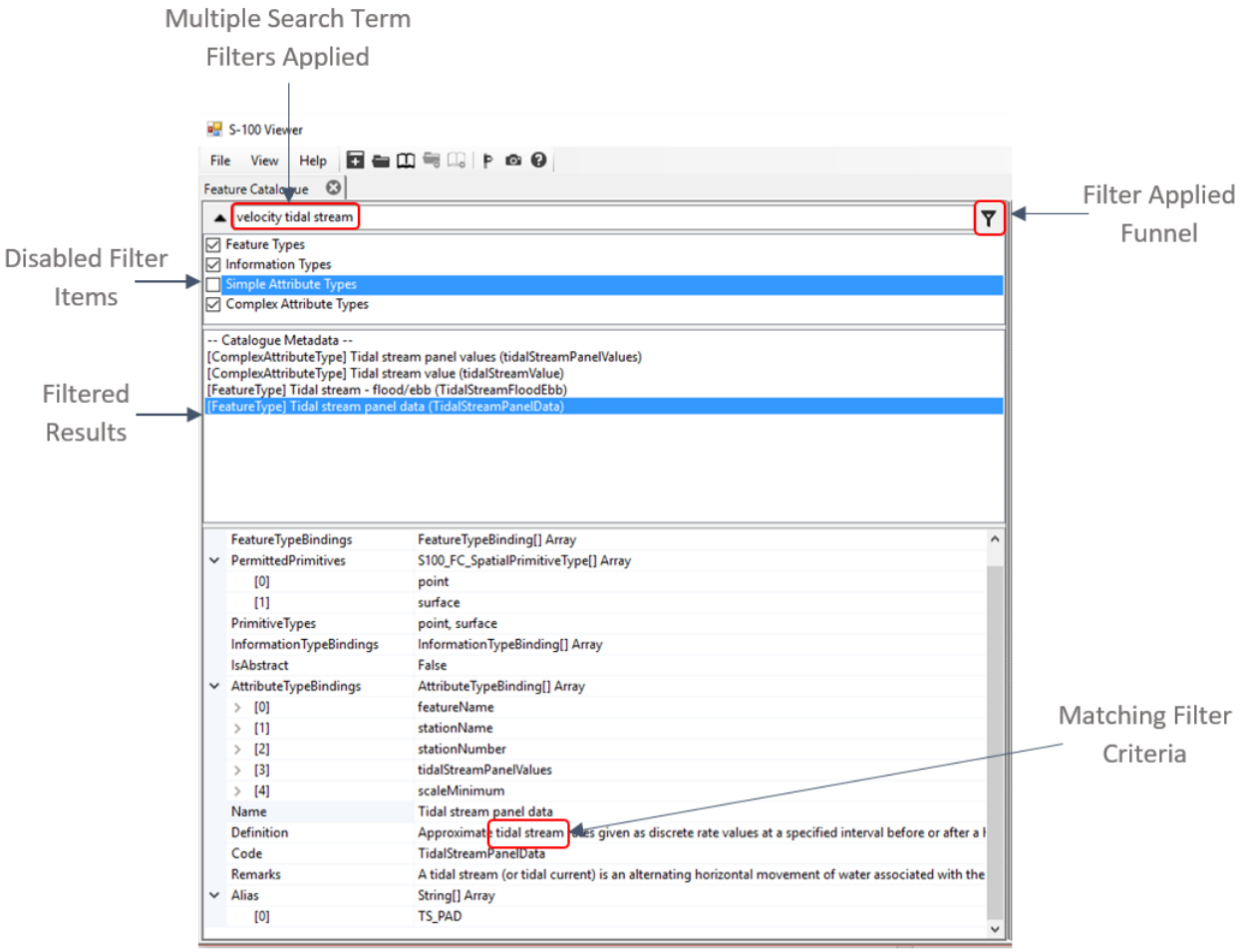


Figure 41. Multiple Filters Applied

5.2.2 Feature Types Entries Block

The “Feature Type Entries Block” shows all feature types that can be reviewed at a given time. By default, with no filter criteria entered in the “Filter Block”, then all defined feature types from the S-100 Feature Catalogue are listed.

5.2.3 Selected Entry Details Block

The “Selected Entry Details Block” shows the associated details for the feature type that has been selected in the “Feature Types Entries Block” (See Figure 41).

5.2.4 Additional Details Block

The “Additional Details Block”, as seen in Figure , offers a panel to further breakdown any selected item in the “Selected Entry Details Block”. Instead of expanding the item inside the “Selected Entry Details Block” the information will be displayed in this block.

6.0 CHART AREA

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The Chart Area is the area where the S-100 data file information is rendered, though currently limited to S-101 data sets. This area allows for visual inspection of the information and relationships defined in the S-100 Feature Catalog, the S-100 Product Catalog, and S-100 data files. The Chart Area consists of the layout seen in Figure .

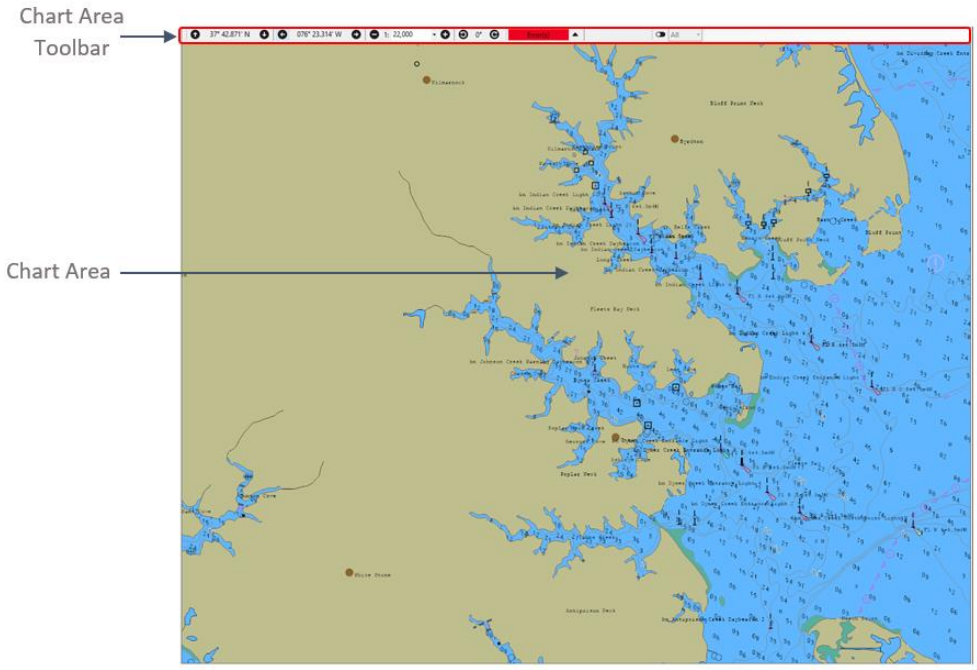


Figure 42. Chart Area Layout

6.1 Chart Area Interaction

To interact with the Chart Area, pan, zoom, and rotate functionality can be implemented through using the Chart Area Toolbar and various hotkeys (See 4.4). Further functionality is also available in the form of spatial filters and feature highlighting.

6.1.1 Spatial Filter

The spatial filter operation provides the ability to select a rectangular portion of the Chart Area to obtain information regarding the internal and overlapping features. The steps to perform a spatial filter include:

1. Identify an object or objects to query
2. Move the cursor to a position near the object in the chart area.
3. Click and hold the left mouse button while dragging the cursor in a diagonal direction across the object. This creates a rectangular region (See Figure) that covers the object.
4. Release the left mouse button when the rectangle covers the object.

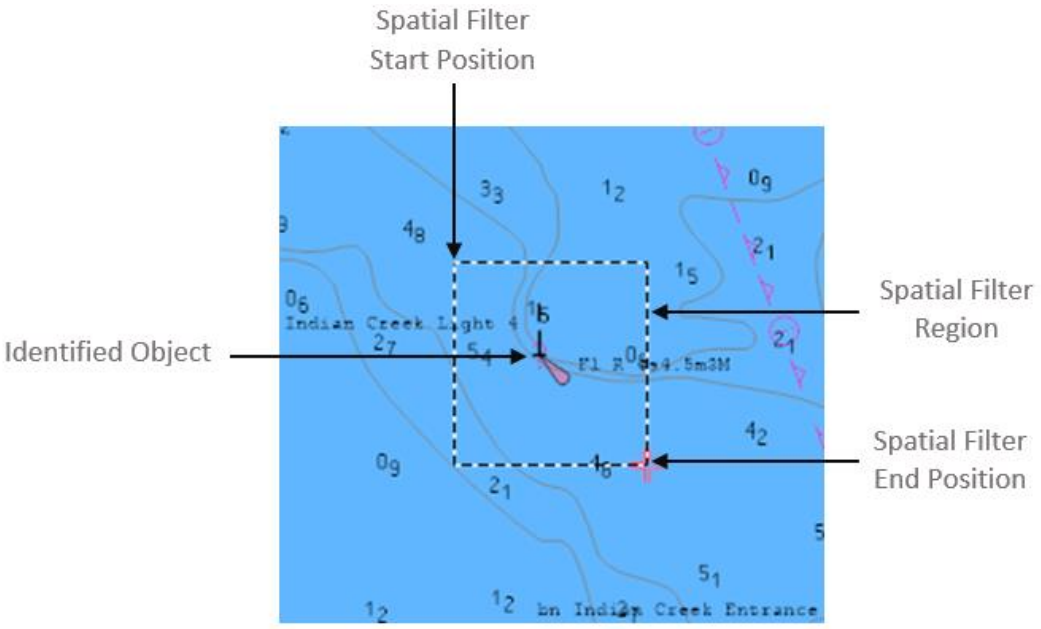


Figure 43. Performing Spatial Filter

Once the mouse button has been released, the Chart Area outside the spatial filter region loses focus while the spatial filter region itself does not. The internal and overlapping features of this filtering region will be populated in the "Feature Record Entries Block" of the S-100 Browser Tab (See Figure).

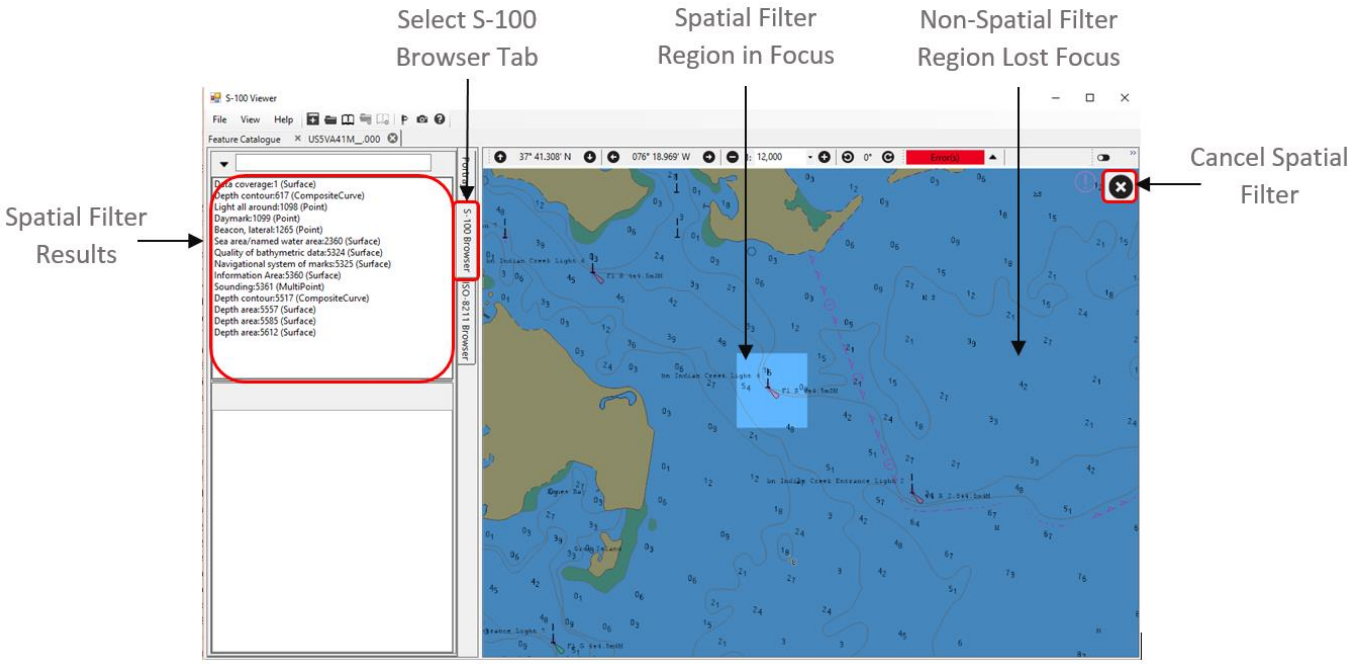


Figure 44. Spatial Filter Results

To cancel the spatial filter, simply click the "Cancel Spatial Filter" button in the top right corner of the Chart Area (See Figure). This button contains a black circle with a "X" inside it.

6.1.2 Feature Highlighting

As features are selected in the "Feature Record Entries Block" of the S-100 Browser Tab (See Figure), they are highlighted in the Chart Area to help identify its location and geometry. Depending on the associated spatials that make up this feature, this highlight can be shown as either a circle that surrounds the feature (See Figure 45) or as a highlight that outlines the geometry of the feature (See Figure).

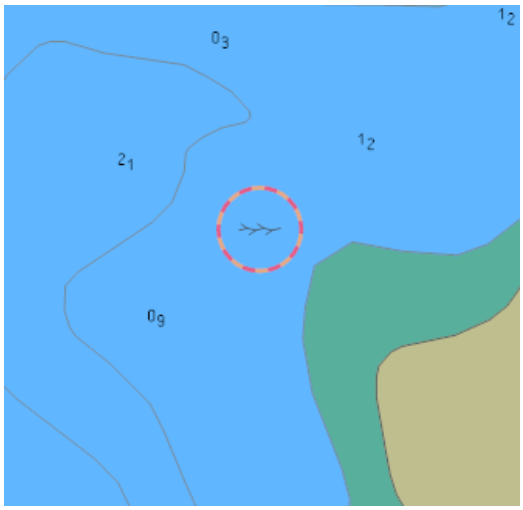


Figure 457. Circled Highlight

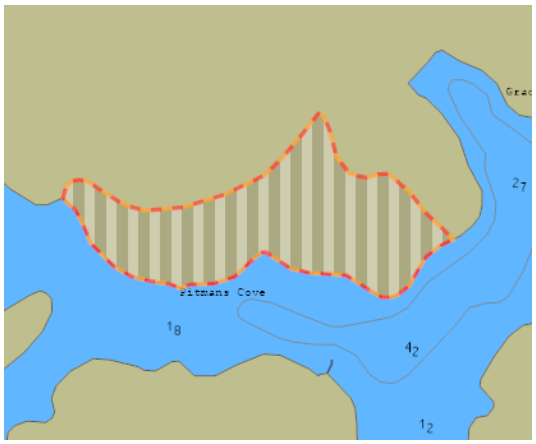


Figure 46. Outlined Highlight

To clear the feature highlight, simply click the "Clear Selection" button in the S-100 Browser Tab (See Figure).

7.0 LOG PANEL

7.0 LOG PANEL

The Log Panel provides a mechanism to display system performance and information about errors or warnings that occur during data set loading or rendering operations. The layout for the Log Panel consists of a Diagnostics and Performance tab as seen in Figure . The panel is hidden by default, so to display it simply click the Hide/Show Log Panel Button on the Chart Area Toolbar identified in Section 4.4, Figure . Each column inside both of the tabs can be sorted forward or reverse by clicking the column header the appropriate number of times.

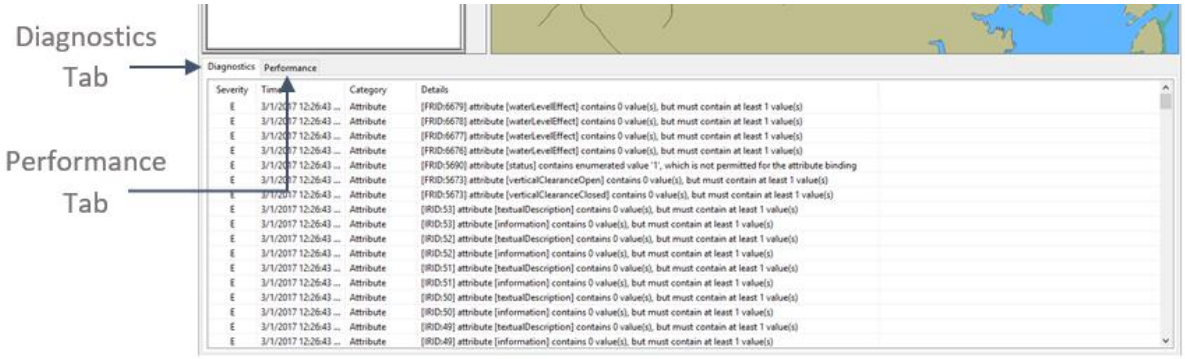


Figure 8. Log Panel Layout

7.1 Diagnostics Tab

The Diagnostics Tab provides an area for review of any encoding or portrayal errors, warnings, or information that occurred upon loading or rendering the data set. This tab is organized into Severity, Time, Category, and Details columns as seen in Figure . Contents of the Diagnostics Tab can be copied and saved by using the mouse to right click inside the area.

Diagnostics | Performance

Severity	Time	Category	Details
E	3/1/2017 12:26:43 PM	Attribute	[FRID:6679] attribute [waterLevelEffect] contains 0 value(s), but must contain at least 1 value(s)
E	3/1/2017 12:26:43 PM	Attribute	[FRID:6678] attribute [waterLevelEffect] contains 0 value(s), but must contain at least 1 value(s)
E	3/1/2017 12:26:43 PM	Attribute	[FRID:6677] attribute [waterLevelEffect] contains 0 value(s), but must contain at least 1 value(s)
E	3/1/2017 12:26:43 PM	Attribute	[FRID:6676] attribute [waterLevelEffect] contains 0 value(s), but must contain at least 1 value(s)
E	3/1/2017 12:26:43 PM	Attribute	[FRID:5690] attribute [status] contains enumerated value '1', which is not permitted for the attribute binding
E	3/1/2017 12:26:43 PM	Attribute	[FRID:5673] attribute [verticalClearanceOpen] contains 0 value(s), but must contain at least 1 value(s)
E	3/1/2017 12:26:43 PM	Attribute	[FRID:5673] attribute [verticalClearanceClosed] contains 0 value(s), but must contain at least 1 value(s)
E	3/1/2017 12:26:43 PM	Attribute	[IRID:53] attribute [textualDescription] contains 0 value(s), but must contain at least 1 value(s)
E	3/1/2017 12:26:43 PM	Attribute	[IRID:53] attribute [information] contains 0 value(s), but must contain at least 1 value(s)
E	3/1/2017 12:26:43 PM	Attribute	[IRID:52] attribute [textualDescription] contains 0 value(s), but must contain at least 1 value(s)
E	3/1/2017 12:26:43 PM	Attribute	[IRID:52] attribute [information] contains 0 value(s), but must contain at least 1 value(s)
E	3/1/2017 12:26:43 PM	Attribute	[IRID:51] attribute [textualDescription] contains 0 value(s), but must contain at least 1 value(s)
E	3/1/2017 12:26:43 PM	Attribute	[IRID:51] attribute [information] contains 0 value(s), but must contain at least 1 value(s)
E	3/1/2017 12:26:43 PM	Attribute	[IRID:50] attribute [textualDescription] contains 0 value(s), but must contain at least 1 value(s)
E	3/1/2017 12:26:43 PM	Attribute	[IRID:50] attribute [information] contains 0 value(s), but must contain at least 1 value(s)
E	3/1/2017 12:26:43 PM	Attribute	[IRID:49] attribute [textualDescription] contains 0 value(s), but must contain at least 1 value(s)

Figure 48. Diagnostics Tab Layout

7.2 Performance Tab

The Performance Tab provides an area for review of detailed timing information ranging from draw instruction generation to rendering operations for applied layers. This tab is organized into Time, Category, and Details columns as seen in Figure 49.

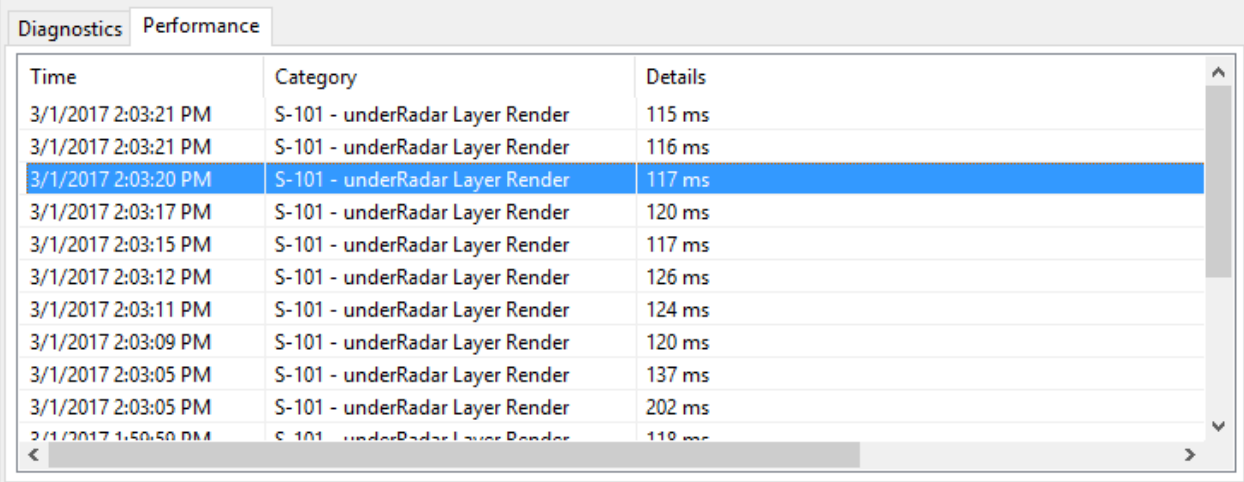


Figure 499. Performance Tab Layout